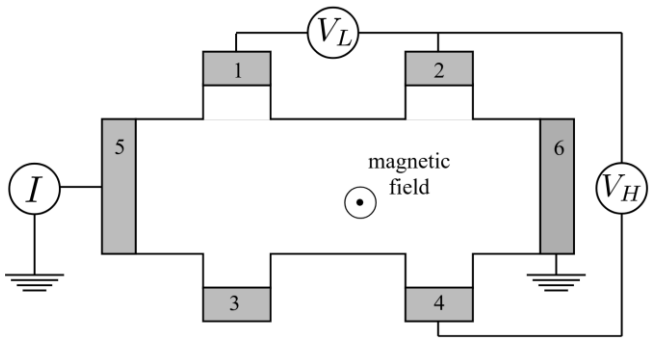
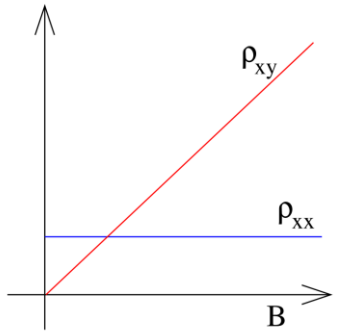
Outline

- Quantum Hall Effect
- Topology and Chern Insulator (CI)
- Haldane Model and CI -- Lattice Version of IQHE
- Fractional Quantum Hall Effect
- Fractional Chern Insulator (FCI) -- Lattice Version of FQAH
- Optical and transport measurements to see FQAH in $MoTe_2$
- SET and optical evidence of QAH in tWSe2

Quantum Hall effect History



Classic Hall effect by drude model

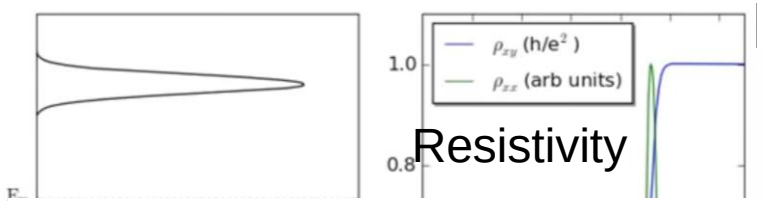


Resistivity

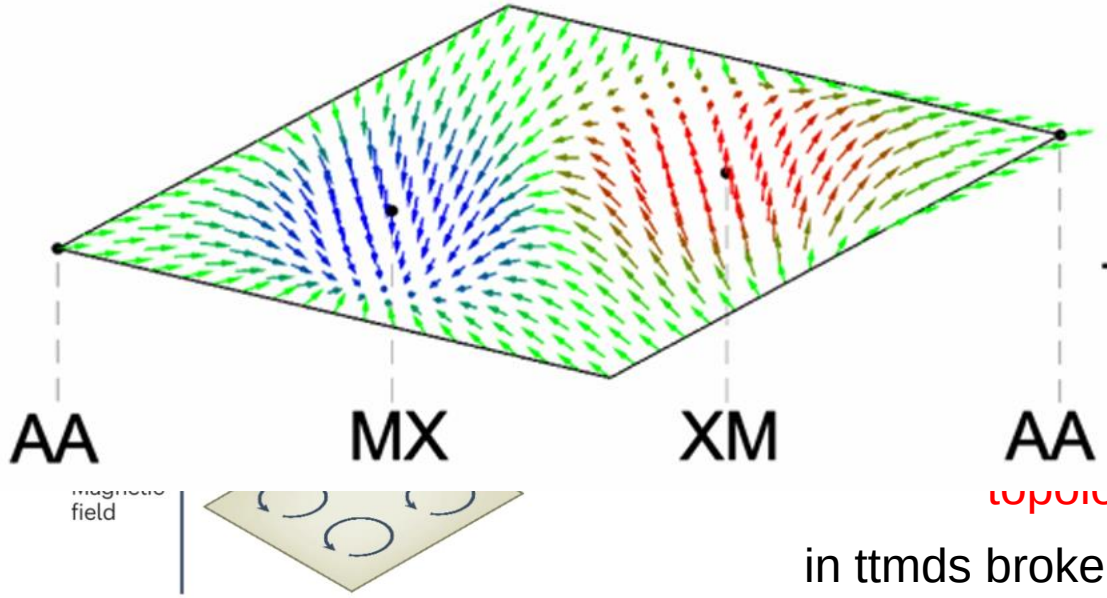
$$\rho_{xx} = \frac{m}{ne^2\tau} \qquad \rho_{xy} = \frac{B}{ne}$$

1. measures the density that does conduct
2. nothing with scattering

Integer quantum Hall effect (in LL)



Resistivity



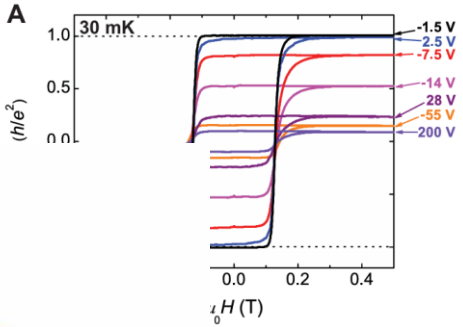
C=1 is given when LL is filled

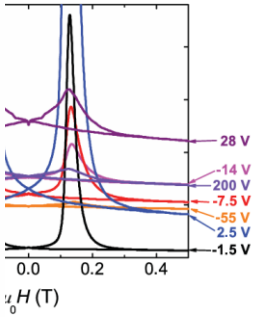
$$\sigma_{xy} = \frac{e^2}{h} \sum_{\text{filled LLs}} C_n = n\frac{e^2}{h}$$

Integer quantum anomalous Hall effect (without LL)

QAHE







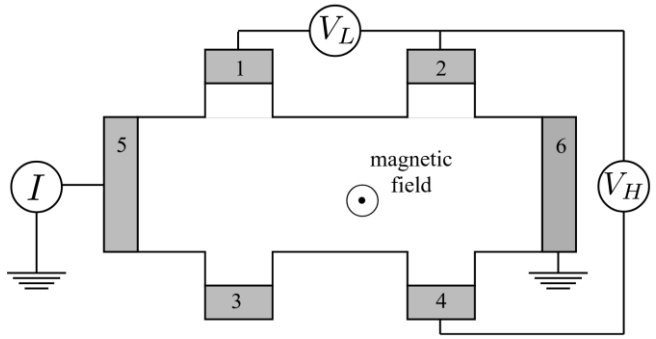
topology and magnetism

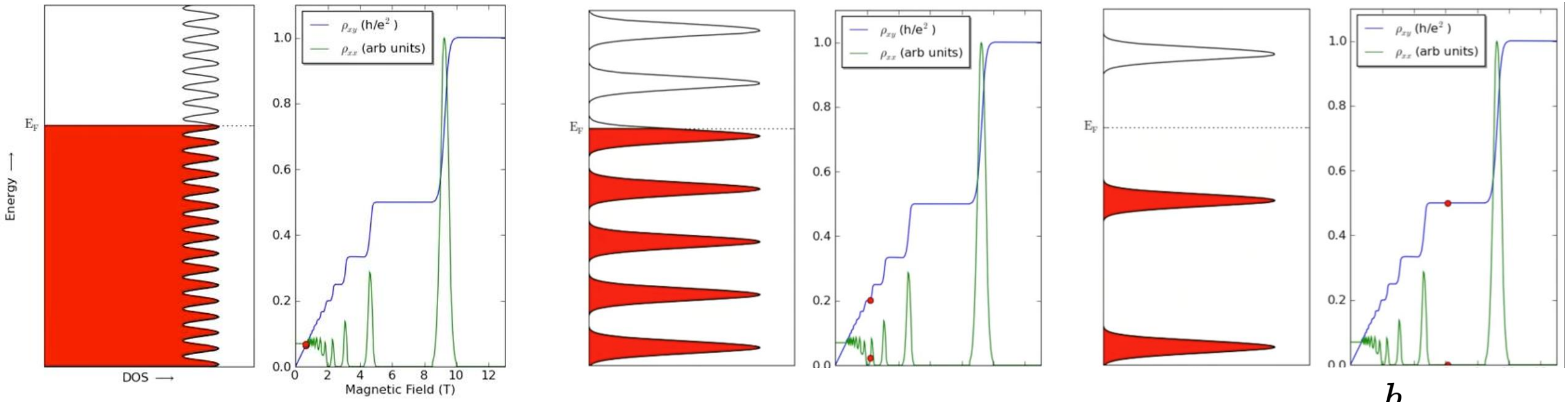
in ttmds broken inversion symmetry -- pseudo spin chirality makes FM happen

$$(\mathbf{S}_i \times \mathbf{S}_j)$$

Cui zu, Chang, Qikun Xue, et al, 2013

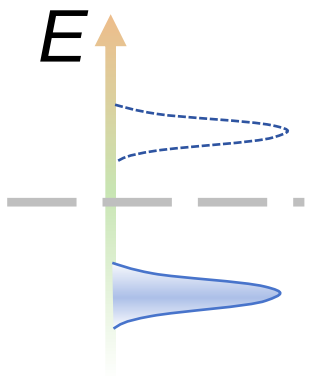
Stages in Quantum Hall effect



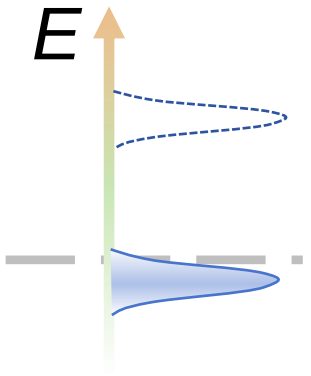


$$\rho_{xy} = \frac{h}{\nu e^2}$$

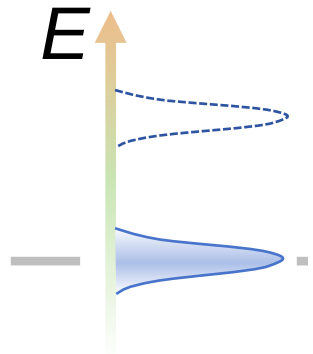
Quantum Hall effect requires disorder (Anderson localization based)



disorders trap e⁻:
Rxx=0,
Rxy plateau



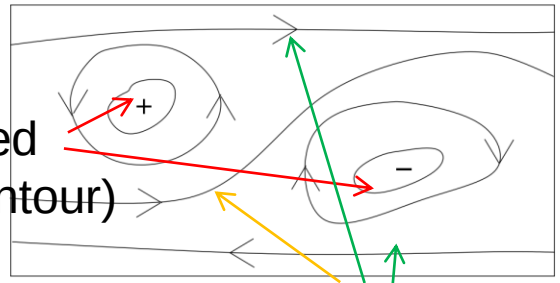
localized state
hosts e⁻:
Rxx=0,
Rxy plateau



extended state:
Rxx>0,
Rxy changing

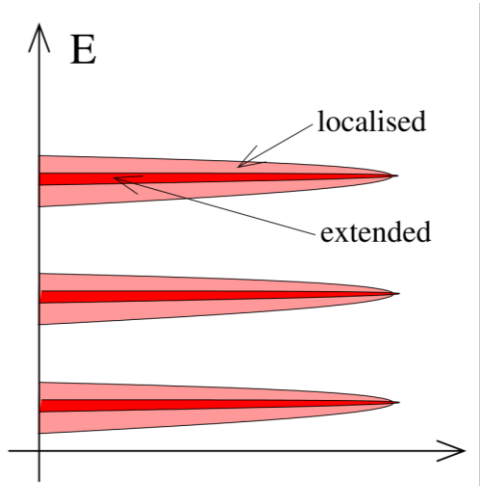
Global: Rxy--charges contributing to the edge

equipotential map



localized
(closed contour)

extended
(percolating contour)



Everytime filling a LL, at extended state:
equipotential contours redistribute,
New percolating contour appears--Rxx
Edge current changes -- Rxy

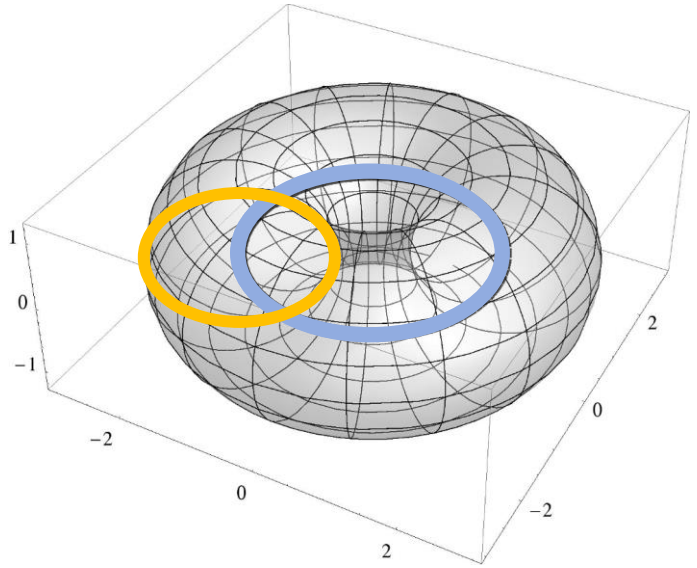
Plateau: Fermi level in localized region (in gap/localized state)

→ bulk insulating
→ no extended states
→ no gap closing
→ Chern is fixed
→ Rxx = 0
→ Rxy plateau (edge current)

Transition point: Fermi level corsses extended state

→ gap closing
→ Chern changes
→ Rxx peak shows up

Berry Phase



how the wavefunction rotates as you move in parameter space

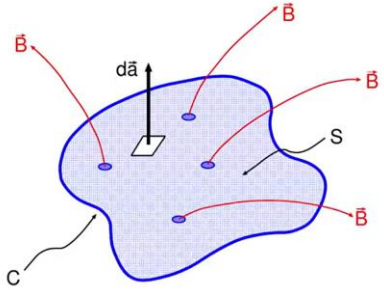
Berry connection

$$\mathbf{A}(k) = i\langle u_k | \nabla_k u_k \rangle$$

the "curl" of that rotation, how many "flux" through surface

Berry curvature

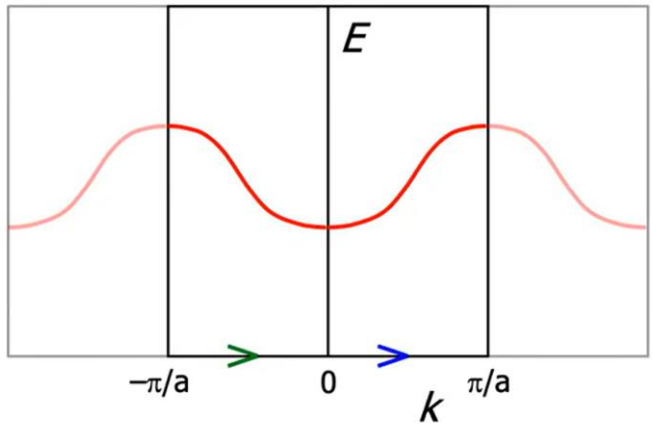
$$\Omega(k) = \nabla_k \times \mathbf{A}(k)$$

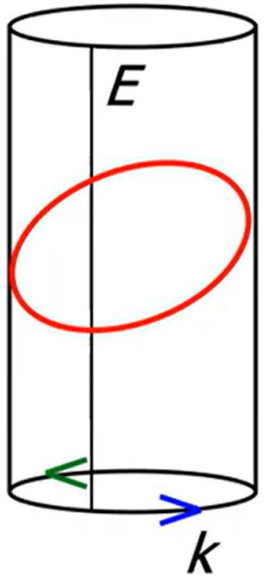


Berry phase

$$\gamma = \oint A(k) \cdot dk = \int \Omega(k)\, dS$$

geometric phase accumulated when the wavefunction travels around a loop





A nontrivial winding of the wavefunction over the Brillouin zone is stable:

$$\int_{BZ} \Omega(k) d^2k = 2\pi C, \quad C \neq 0$$

Symmetry and case in LLs

$$\int_{BZ} \Omega(k) d^2k = 2\pi C, \quad C \neq 0$$

IS: $\Omega(k) = \Omega(-k)$

TRS $\Omega(k) = -\Omega(-k)$

Nonmag. and Centrosymm.:

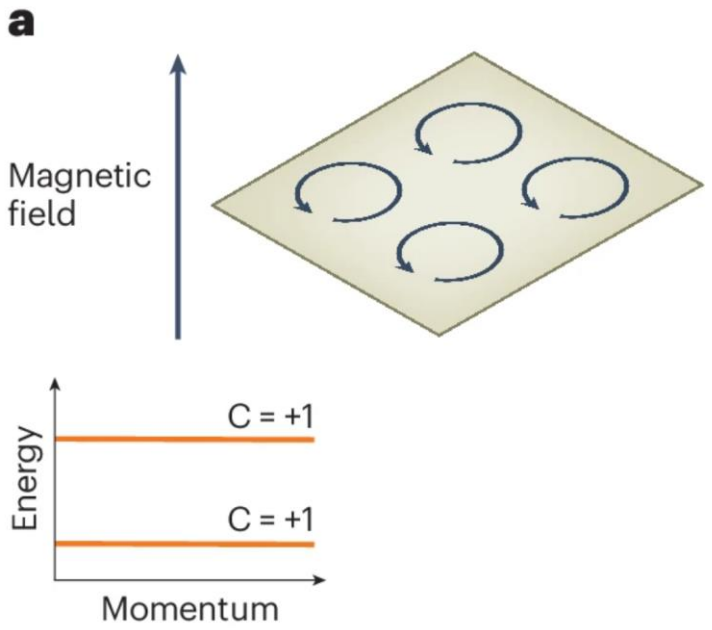
$$\Omega(k) = 0$$

Nonmag. and non-centrosymm.:

$$\int_{BZ} \Omega(k) d^2k = 0$$

With B field or FMI (TR broken):

$$\int_{BZ} \Omega(k) d^2k \neq 0$$



Berry curvature of a Landau level is completely uniform:

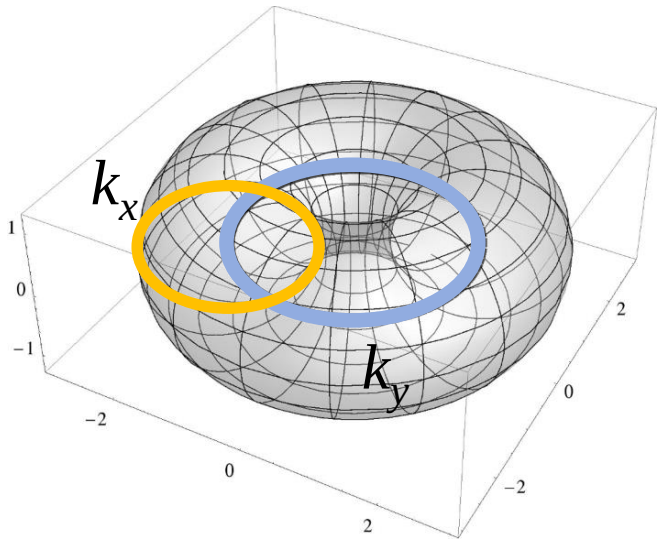
$$\Omega = \frac{1}{\ell_B^2}$$

$$l_B = \sqrt{\hbar/(eB)}$$

The Chern number counts how many flux quanta penetrate for a filled LL

$$C = 1 \quad \text{per LL}$$

Topology and in the Chern Insulator



C from torus T^2

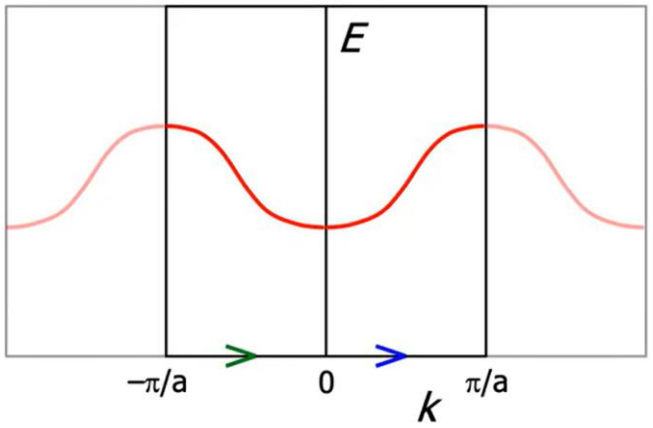
$$\sigma_{xy} = \frac{e^2}{h} \sum_{\text{filled LLs}} C_n = n\frac{e^2}{h}$$

every LL carries a topological unit (C=1)

$$\Omega(\mathbf{k}) = \frac{1}{(l_B)^2} = B$$

$$C = \frac{1}{2\pi} \int d^2k\, \Omega(k) = 1$$

For a square lattice,



inversiion symmetry

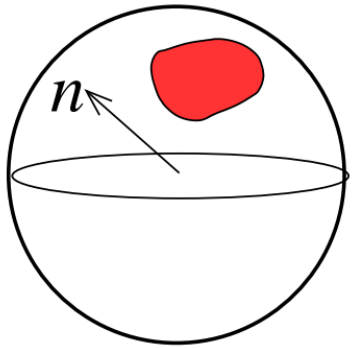
$$(\sin k_y)\sigma_2 + (m + \cos k_x + \cos k_y)\sigma_3$$

< 0

< 2

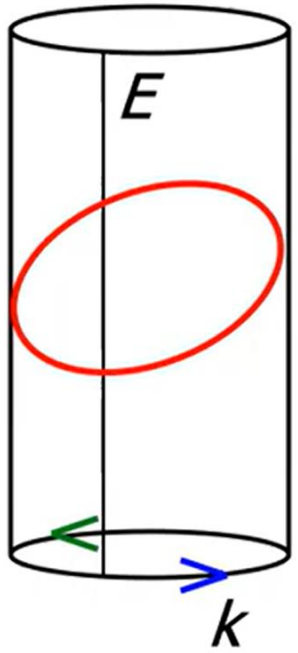
2

$$) = \frac{\vec{E}(\mathbf{k})}{|\vec{E}(\mathbf{k})|}$$



Bloch sphere

T^2--S^2



ıly with gap

$\times \partial_{k_y} \mathbf{n})$

$\times \partial_{k_y} \mathbf{n})\, d^2k$

ıs the sphere

C from Brillouin zone

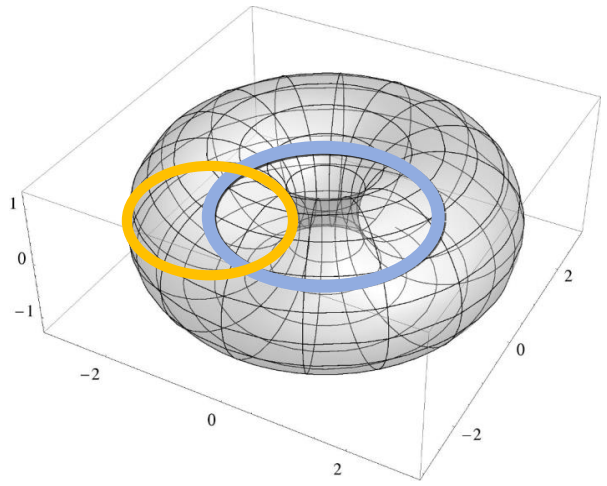
"Effective" B in IQHE without B with a non-zero Chern number!

$$\Omega(k) = \nabla_k \times A(k)$$

$$C = \frac{1}{2\pi} \int_{\text{BZ}} \Omega(\mathbf{k})\, d^2k$$

$$\sigma_{xy} = \frac{e^2}{h} C$$

Haldane model --topology is the key for integer quantized hall effect

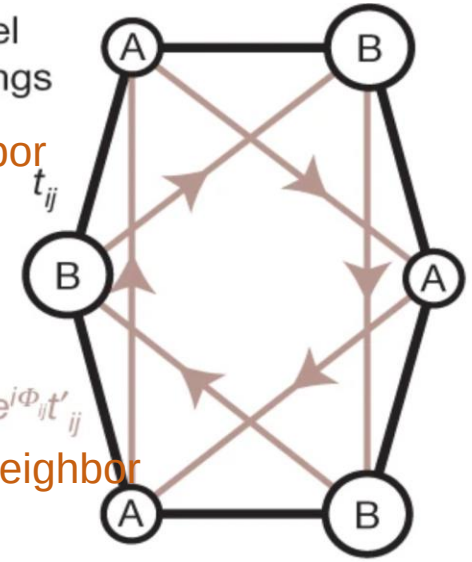


Tunnel couplings

Nearest-neighbor hopping (NN) t_{ij}

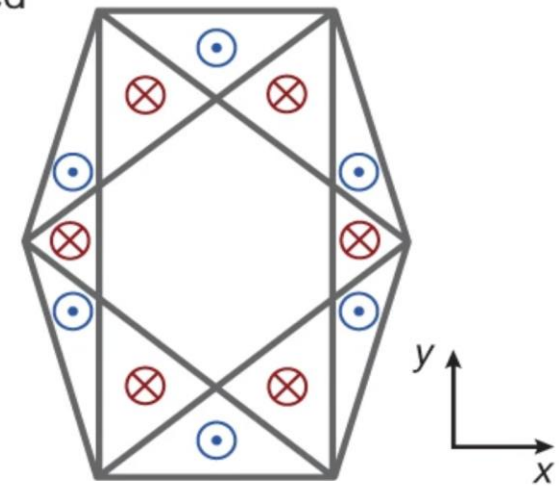
$e^{i\Phi_{ij}}t'_{ij}$

Next-nearest-neighbor hopping (NNN)



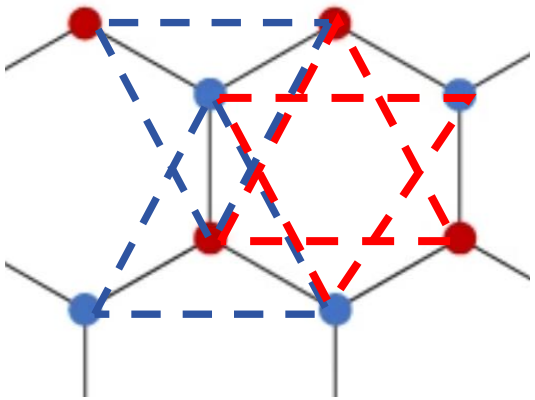


Staggered flux



$$\sum_{\text{hexagon}} \Phi = 0,$$

$\Phi \neq 0$ on individual loops



C from torus T^2

$$\sigma_{xy} = \frac{e^2}{h} \sum_{\text{filled LLs}} C_n = n\frac{e^2}{h}$$

every LL carries a topological unit (C=1)

$$\Omega(\mathbf{k}) = \frac{1}{(l_B)^2} = B$$

$$C = \frac{1}{2\pi} \int d^2k\, \Omega(k) = 1$$

C from Brillouin zone

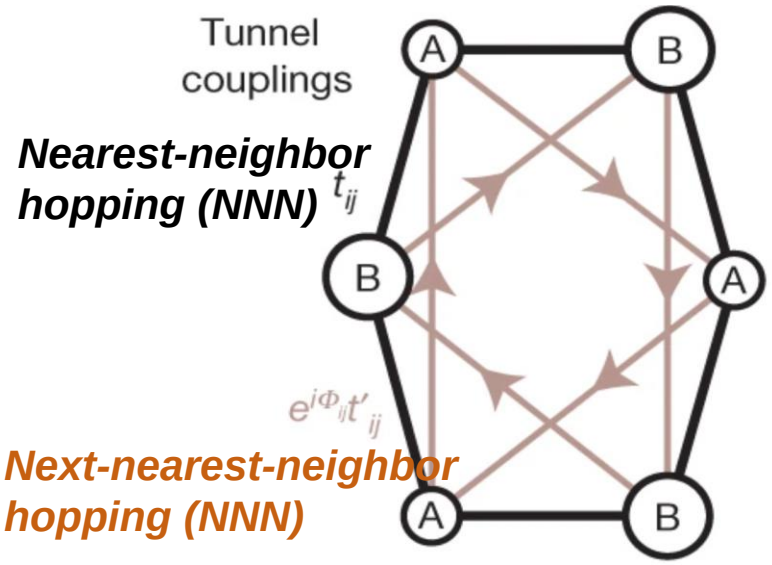
"Effective" B in IQHE without B: Berry curvature + Chern number!

$$\Omega(k) = \nabla_k \times A(k)$$

$$C = \frac{1}{2\pi} \int_{\text{BZ}} \Omega(\mathbf{k})\, d^2k$$

$$\sigma_{xy} = \frac{e^2}{h} C$$

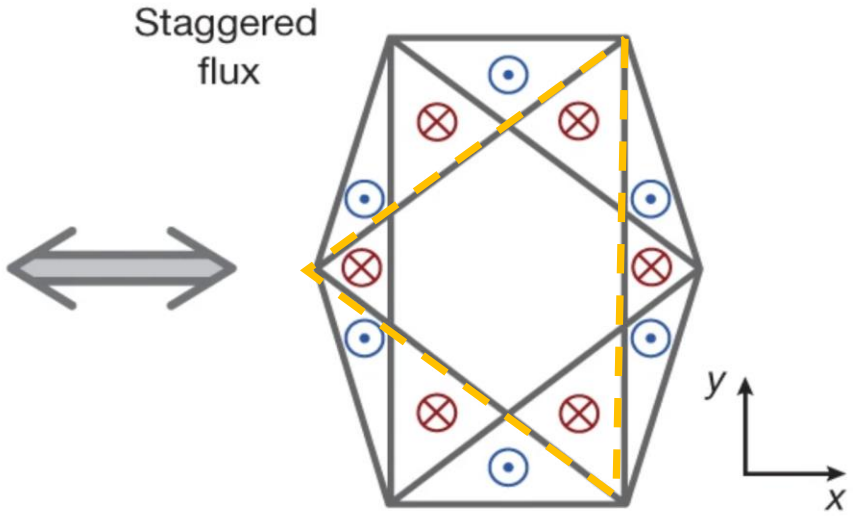
Haldane model --topology is the key for integer quantized hall effect



Nearest-neighbor hopping (NNN) t_{ij}

$e^{i\Phi_{ij}}t'_{ij}$

Next-nearest-neighbor hopping (NNN)



Peierls phase

$$t_{ij} \rightarrow t_{ij}e^{i\theta_{ij}}$$

$$\theta_{ij} = \frac{e}{\hbar}\int_j^i \mathbf{A}\cdot d\mathbf{l}$$

$$\prod_{\text{loop}} e^{i\theta_{ij}} = e^{i\frac{e}{\hbar}\oint \mathbf{A}\cdot d\mathbf{l}} = e^{i\frac{e}{\hbar}\Phi_{\text{loop}}}$$

In triangular lattce:

$$e^{i\Theta_{\triangle}} = e^{i(\theta_{12}+\theta_{23}+\theta_{31})}$$

“Effective” flux:

$$\oint \mathbf{A}\cdot d\mathbf{l} = \iint \mathbf{B}\cdot d\mathbf{S} = \Phi_{\triangle}$$

$$\Theta_{\triangle} = \frac{e}{\hbar}\Phi_{\triangle}$$

Every NNN :

$$\Phi_{\triangle} = \frac{3\phi}{2\pi}\Phi_0$$

Onsite energy in sublattice A&B (broken inversion symmetry):

$$H_M = M\sum_i \xi_i c_i^{\dagger}c_i$$

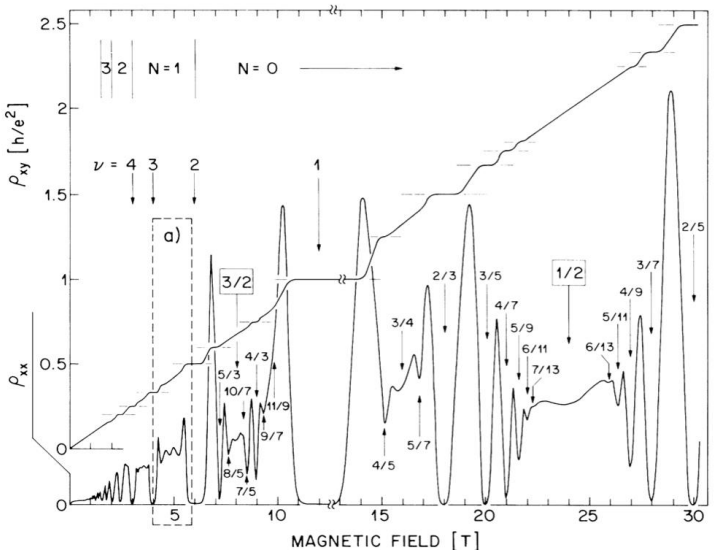
$$\xi_i = \begin{cases} +1, & i \in A \\ -1, & i \in B \end{cases}$$

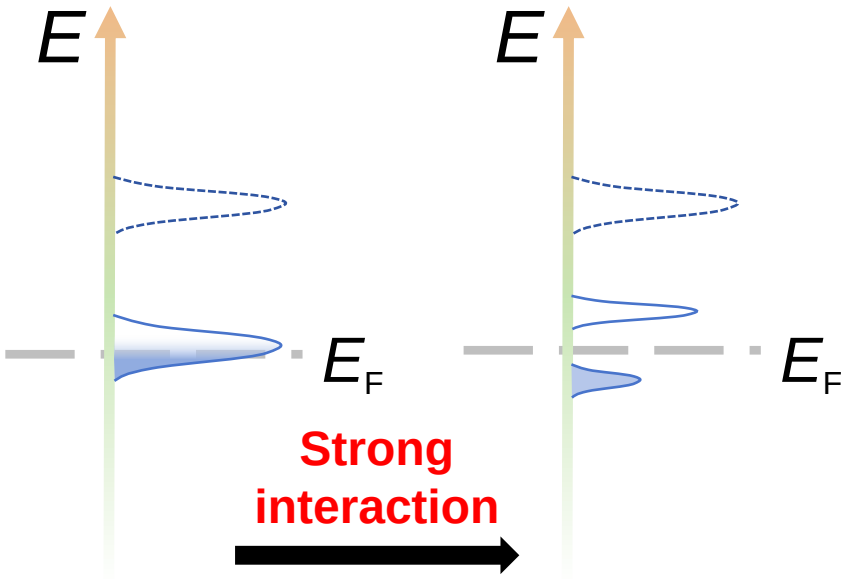
NNN in sublattice A&B, where ferromagnetism from (broken time reversal symmetry):

$$T : e^{i\phi} \not\Rightarrow e^{-i\phi}$$

$$\phi \neq 0, \pi$$

Fractional Quantum Hall states





Laughlin States $\nu = \frac{1}{m} \quad (m = 1, 3, 5, \dots)$

N particle wavefunction:

$$\Psi_m(z_1, \dots, z_N) = \prod_{i<j} (z_i - z_j)^m \exp\left(-\sum_{i=1}^{N} \frac{|z_i|^2}{4l_B^2}\right)$$

Jastrow term

Gaussian term (characteristic width)

$$(z_i - z_j)^m \longrightarrow 0 \quad \text{as } z_i \to z_j$$

Interaction:the probability of finding two electrons close is strongly suppressed

The magnetic length :

$$l_B = \sqrt{\hbar/(eB)}$$

distance between e-s

$$d \sim \sqrt{m}\, l_B$$

Physics in m:

1. how fast the probability vanishes

$$\text{Pauli: } 1$$

$$\text{Laughlin: } m = 3, 5, 7, \dots$$

2. fractional charge estimated by total angulart momenmtum mN^2

$$R \sim \sqrt{2mN}\, l_B$$

$$\rho = \frac{1}{2\pi m l_B^2} \Rightarrow \nu = \frac{1}{m}$$

3. $U_{coulomb}$ (2D plasma)

$$|\Psi_m|^2 = e^{-\beta U(\{z_i\})}$$

$$-m^2 \ln|z_i - z_j|$$

FQAHE

IQHE: Topology
FQHE: Strong interaction+Topology
QAHE: Magnetism+Topology
FQAHE: everything above?

To replace/simulate LLs

degeneracy $N_{\text{degeneracy}} = \dfrac{AB}{\Phi_0}$

quenched kinetic energy

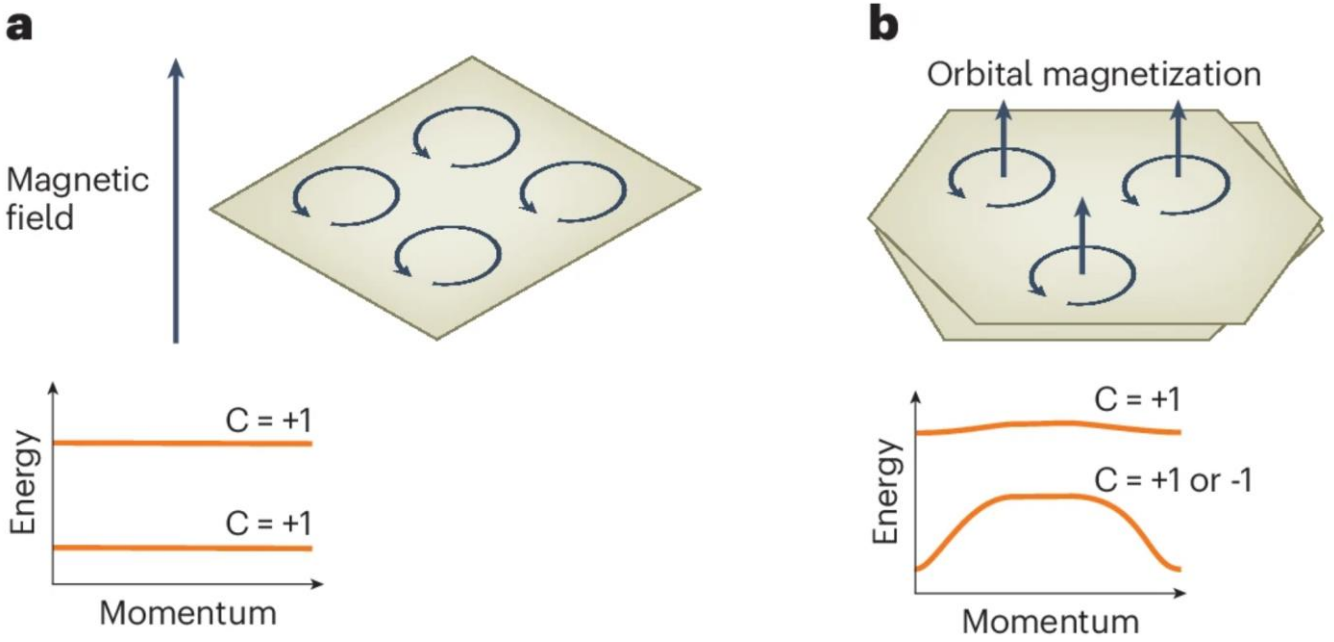
Non hopping between LLs v.s. Flat band:

$$\Delta_{\text{band}} \gg E_{\text{int}}$$

topology $\sigma_{xy} = \dfrac{e^2}{h} \sum_{\text{filled LLs}} C_n = n\dfrac{e^2}{h}$

Morié+Chern Band

Landau level（continuum, C=1）
↔ moiré Chern flat band（lattice, C≠0）

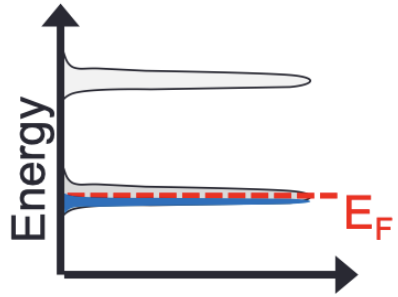


$\dfrac{\Phi}{\Phi_0} = \dfrac{p}{q}$ p effective flux channels, q degeneracy

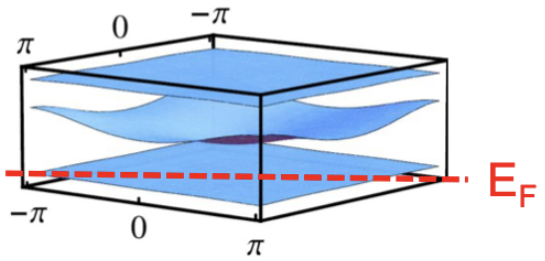
Nat. Phys. 2025, Kenji Yasuda

FCI(FQAH) from FQH

Macroscopic degeneracy:

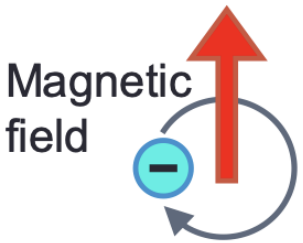


Fractionally filled Landau level



Fractionally filled band

Time-reversal symmetry breaking:





Berry curvature

$$F_{xy} = -i\left[\langle\partial_{k_x}\varphi|\partial_{k_y}\varphi\rangle - \langle\partial_{k_y}\varphi|\partial_{k_x}\varphi\rangle\right]$$

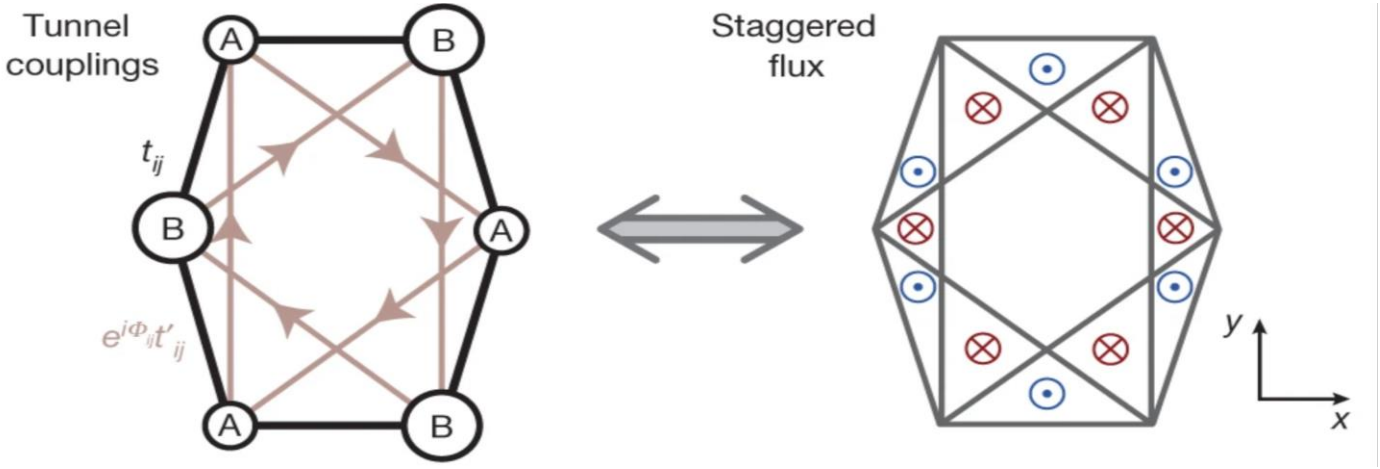
Interaction:

Coulomb $\dfrac{ke^2}{r}$

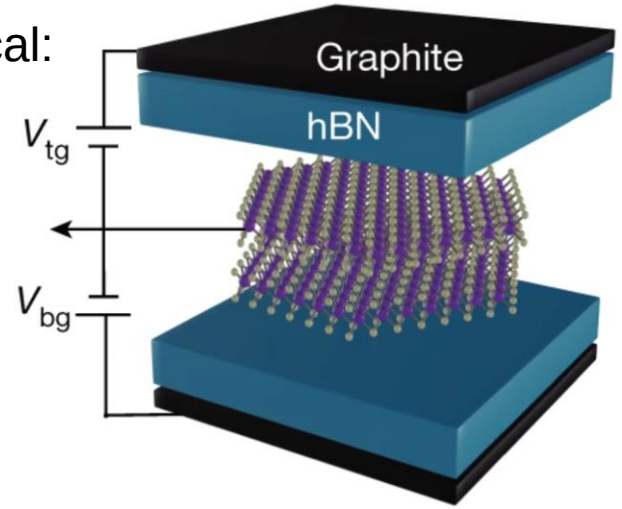
$\sum_{\langle ij\rangle} n_i n_j + \ldots$ Hubbard, etc

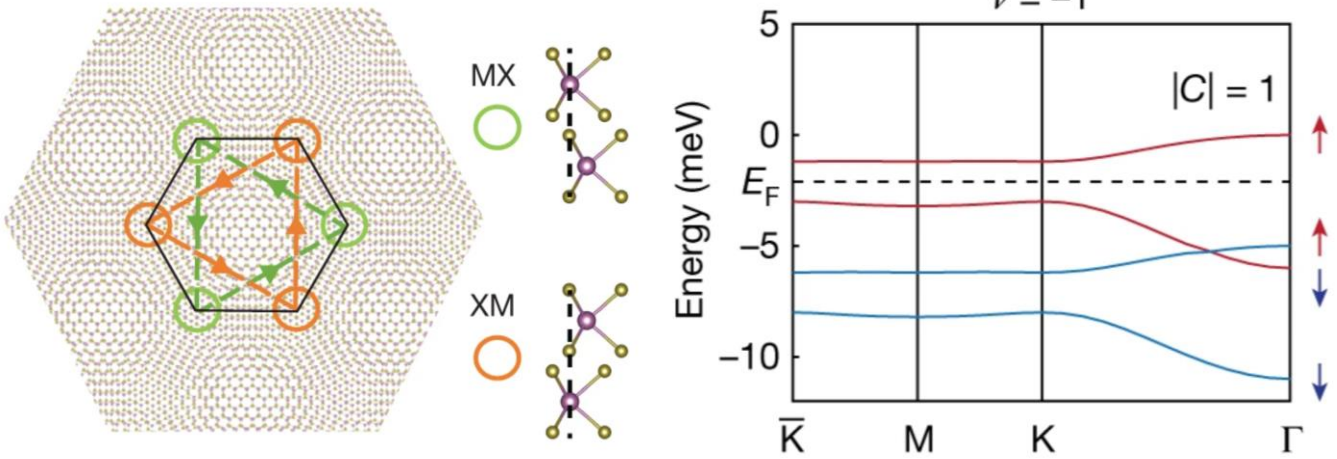
Fractional Quantum Hall **Fractional Chern Insulator**

Reveiling the FQAHE in R-stacked $tMoTe_2$

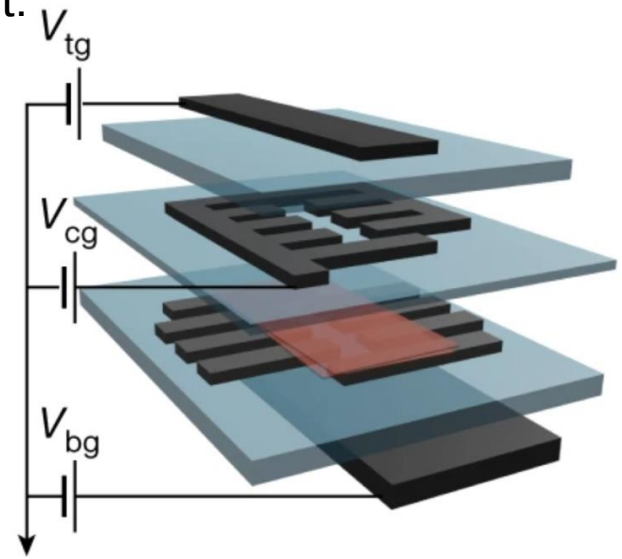


Optical:





Transport:



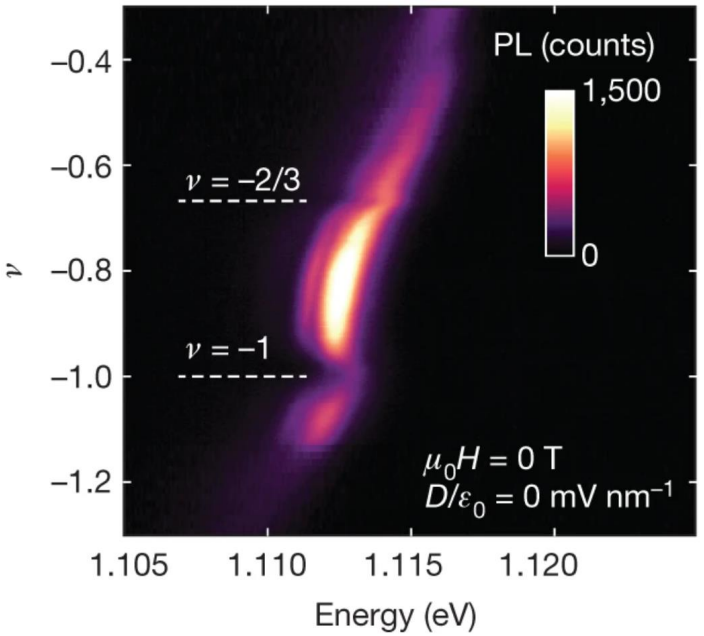
Haldane/Kane-mele -- “effective B”/Anomalous+Chern

Reveiling the FQAHE in R-stacked $tMoTe_2$

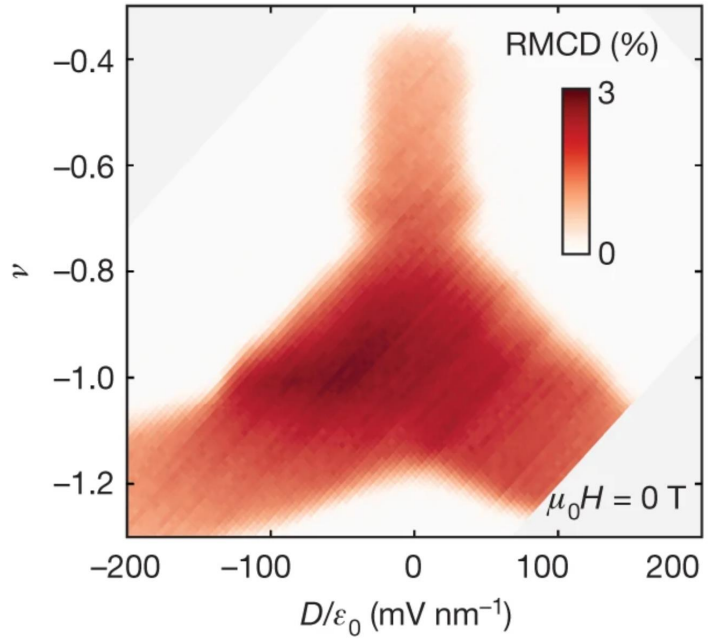
IQHE: Topology
FQHE: Strong interaction+Topology
QAHE: Magnetism+Topology
FQAHE: Everything above?

1. PL: Fractional correlated states
2. RMCD: Magnetism
3. B Field dependence: Topology

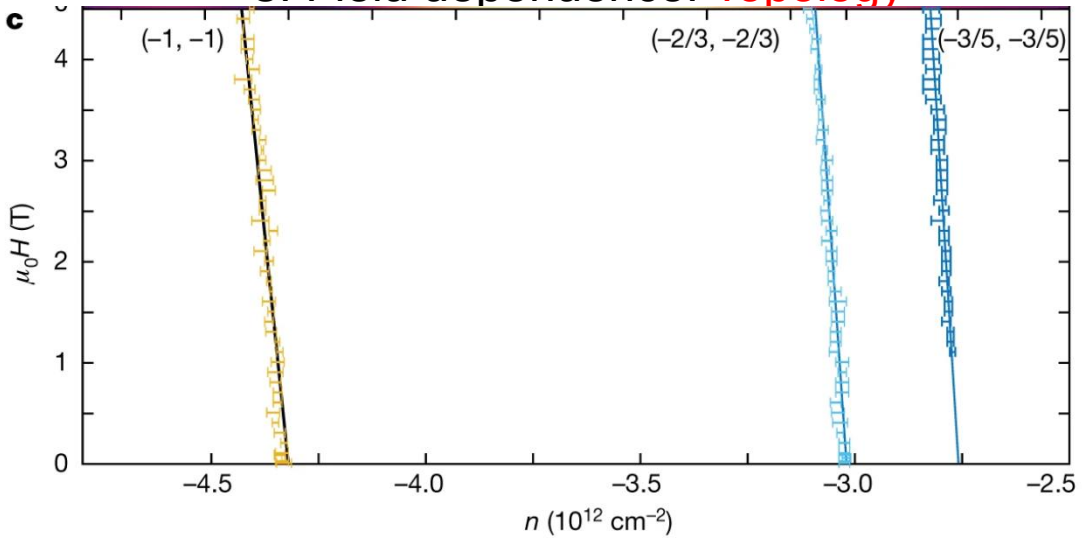
1. Fractional correlated states



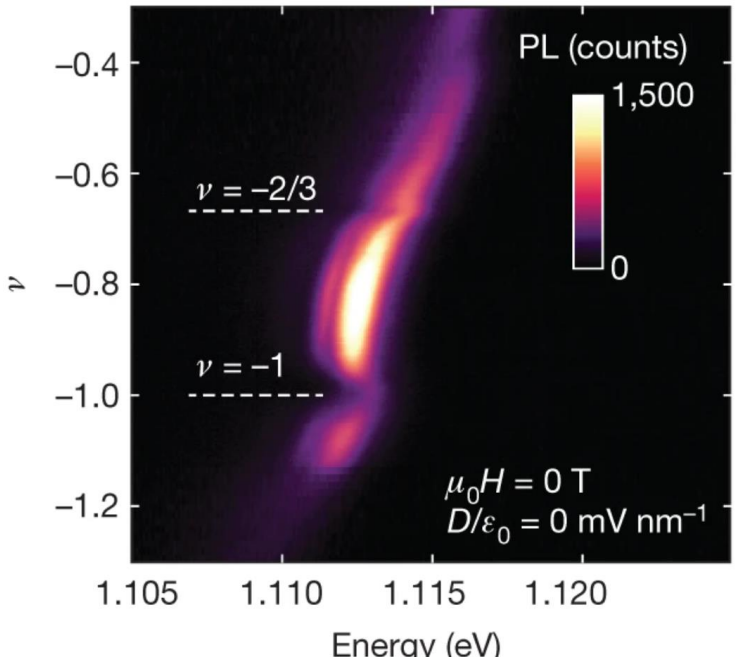
2. RMCD: Magnetism

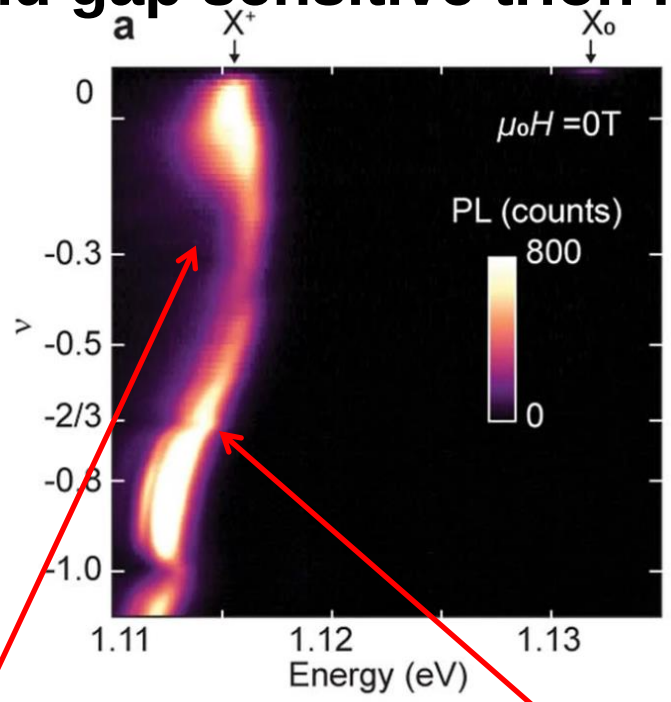


3. Field dependence: Topology

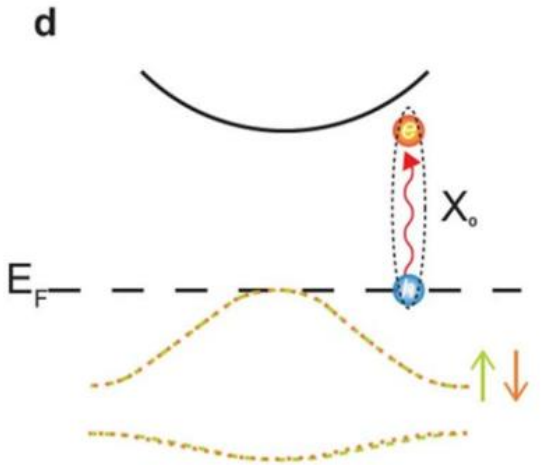


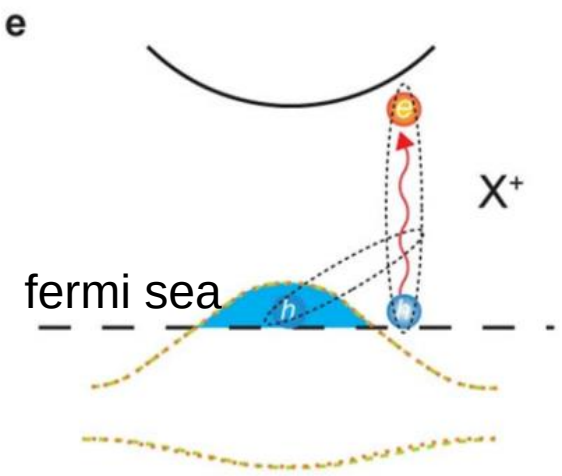
Fractional correlated states and gap sensitive trion PL

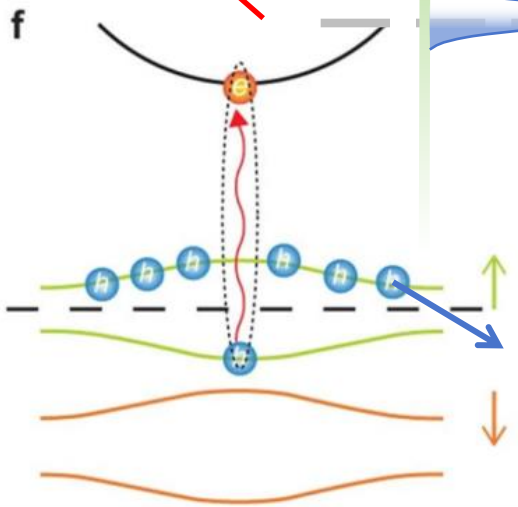


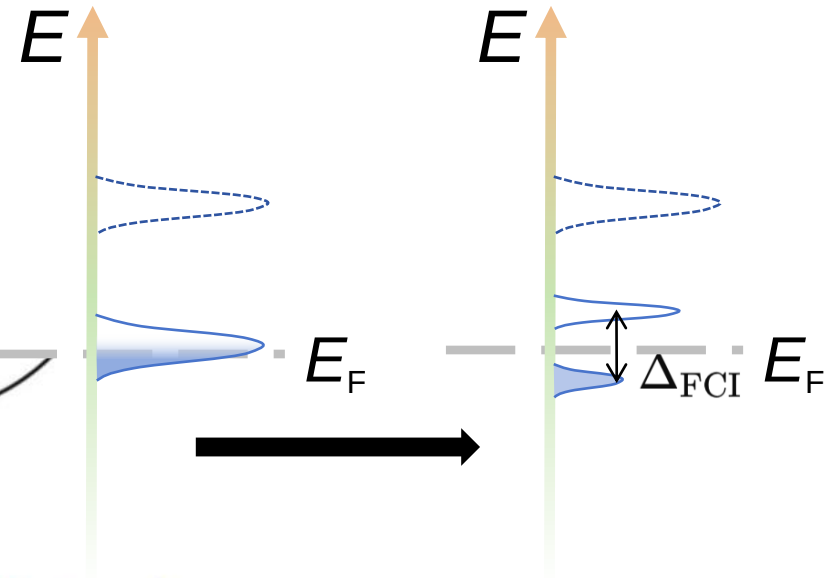


from e-f, at fractional filling, a topological gap open (Allowed states excitations are renormalized):



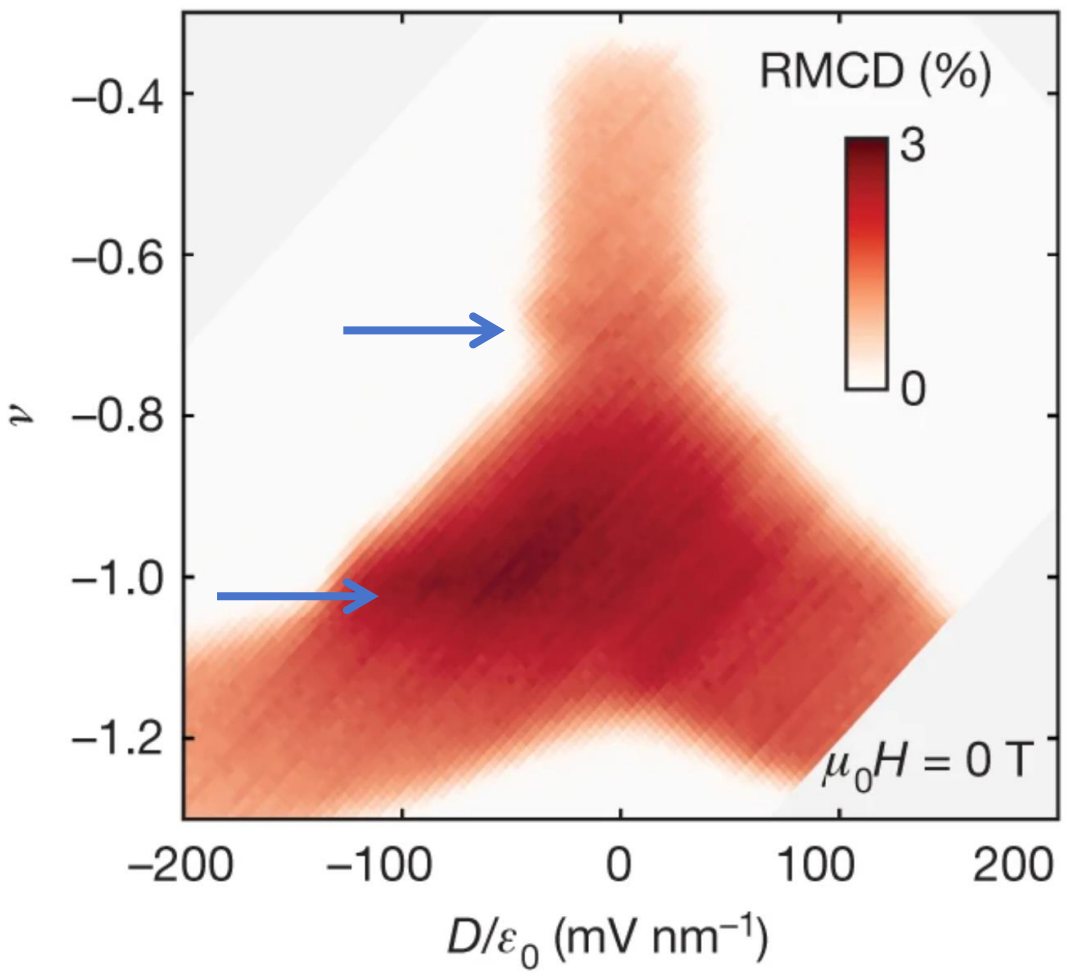




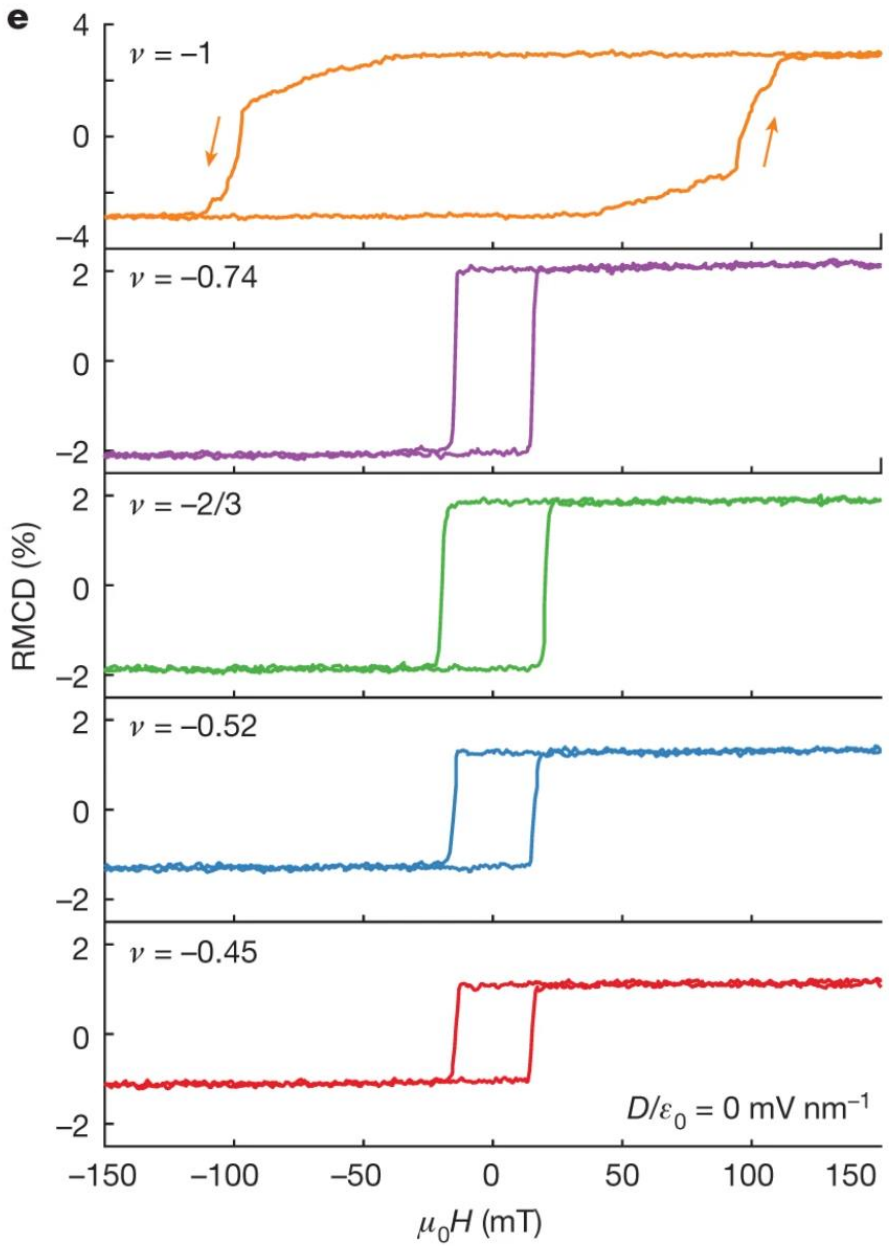


Laughlin-like fractional state

Correlated Magnetic states

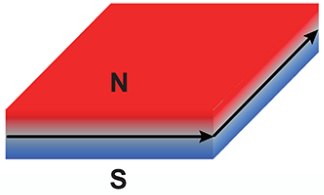


$$\mathrm{RMCD} = \frac{R_{\sigma^+} - R_{\sigma^-}}{R_{\sigma^+} + R_{\sigma^-}}$$



Topological states, FCI--lattice version of FAQHE

Edge current generates a bulk magnetism:



Thermodynamics argument
--Optically detected Fan/ Streda's formula:

$$\sigma_{xy} = ec\left(\frac{\partial N}{\partial B}\right)_{\mu,T}$$

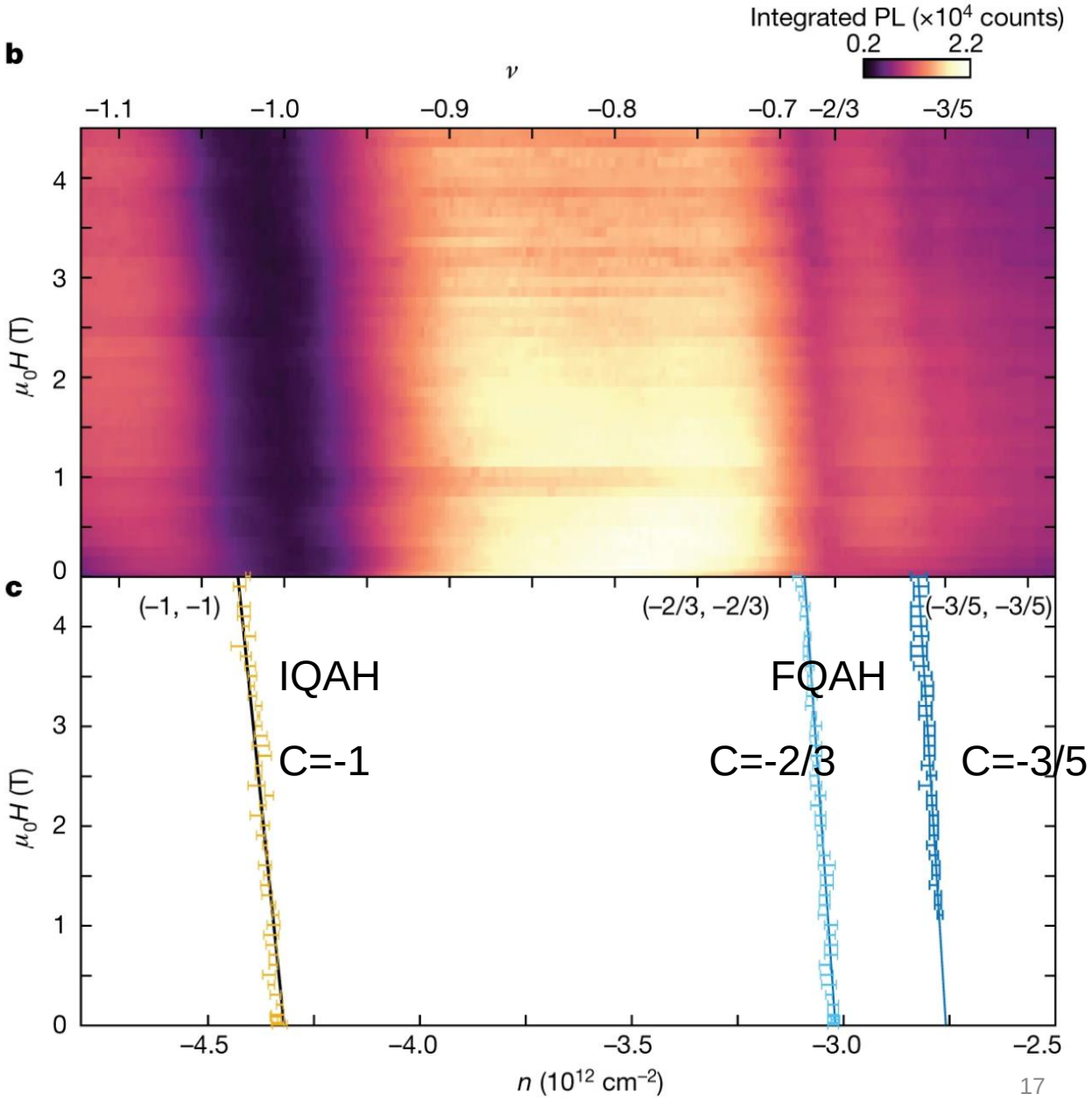
For quantum hall states:

$$N_\phi = \frac{BA}{\Phi_0} \qquad \frac{\partial N}{\partial B} = \frac{A}{\Phi_0}\nu$$

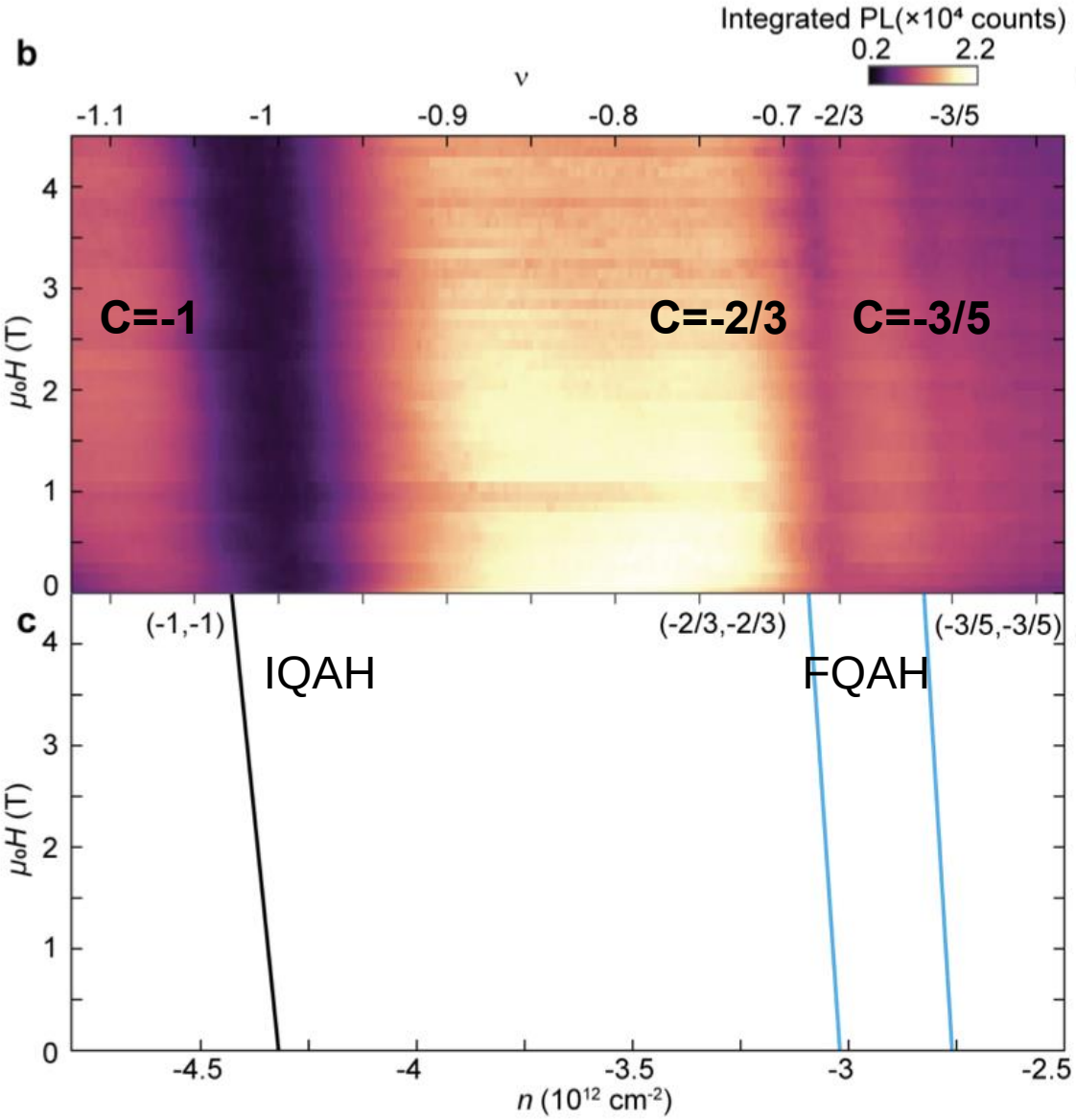
$$\sigma_{xy} = ec \cdot \frac{A}{\Phi_0}\nu \qquad \sigma_{xy} = \frac{I}{E_x A} = \nu\frac{e^2}{h}$$

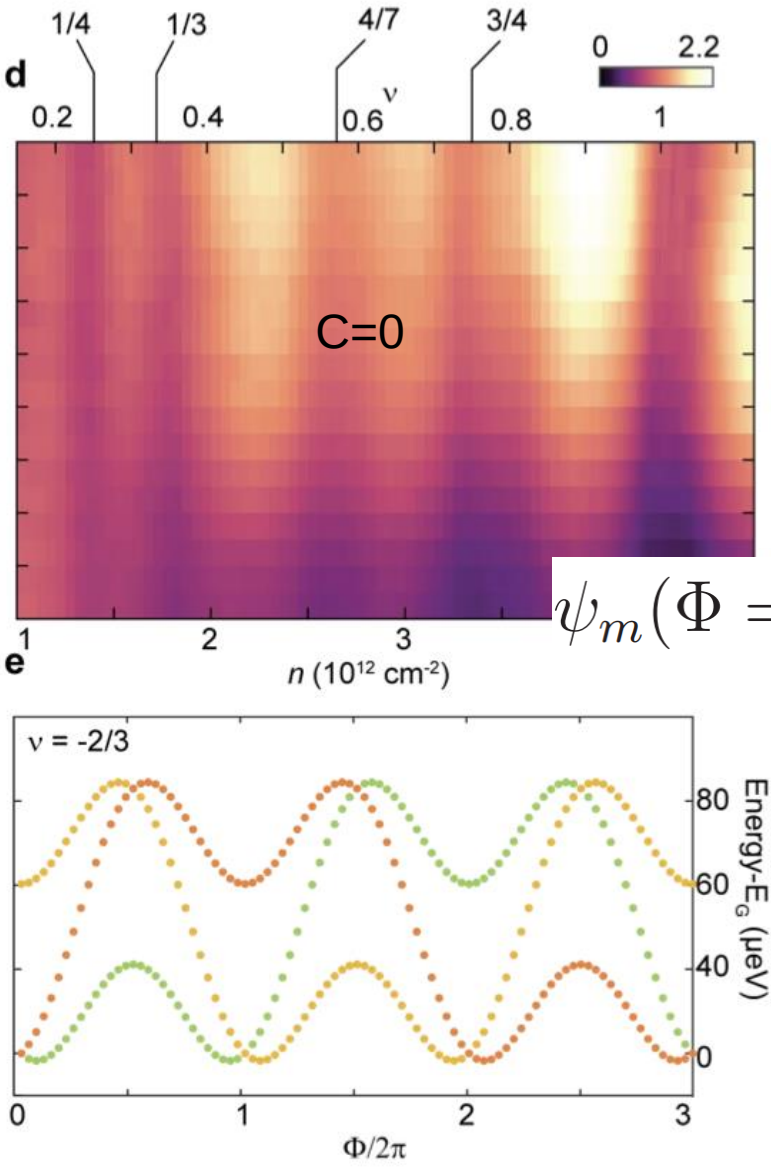
$$C = N_{\text{edge}} = \nu$$

$$C = \phi_0 \frac{\partial n}{\partial B}$$



Topological trivial correlated states

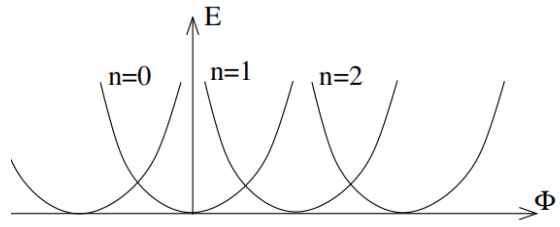




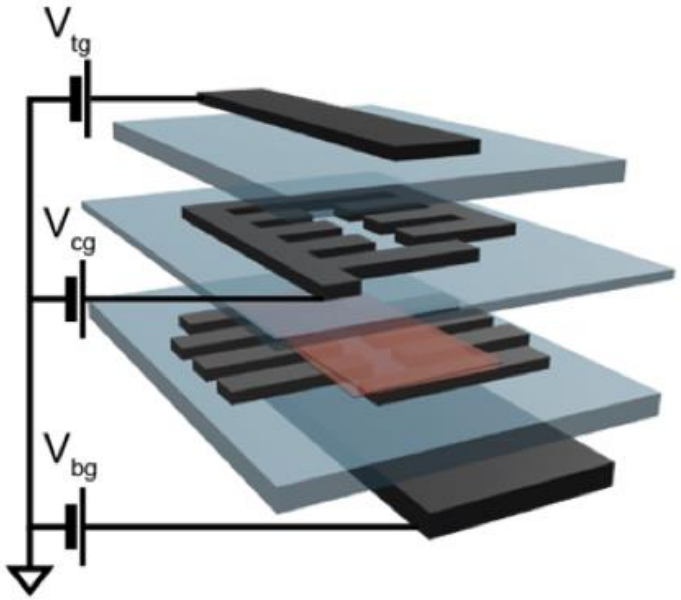
$$\psi_m(\Phi = 0) \longrightarrow$$

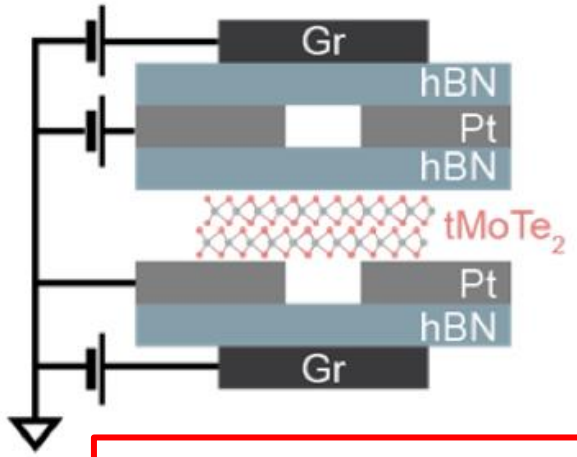
$$\psi_m(\Phi = \Phi_0) = \psi_{m+1}(\Phi = 0$$

sptectral flow



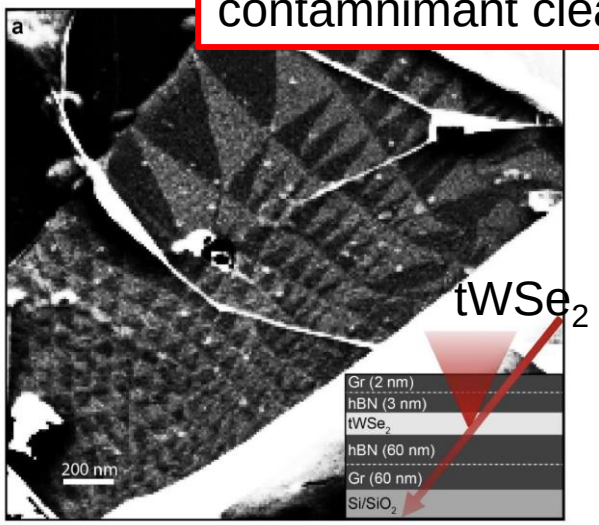
Transport device to confirm FQAH in $tMoTe_2$

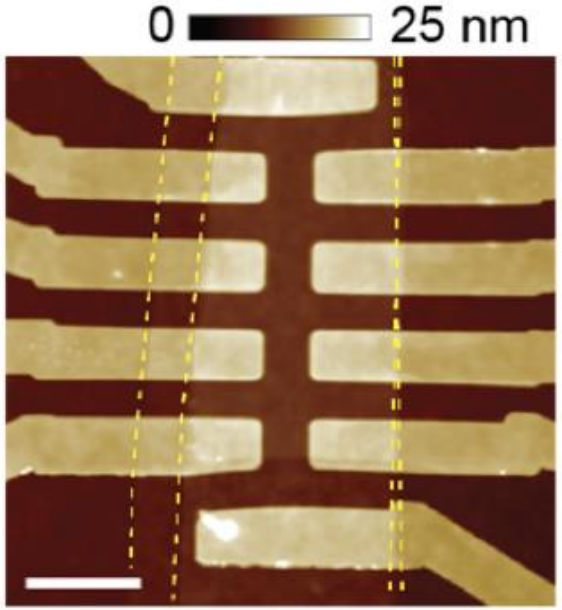




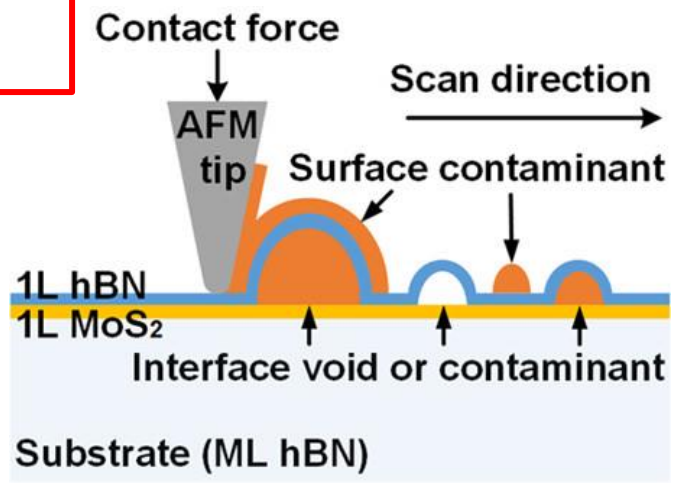
1. Contact engineering to overcome SHB

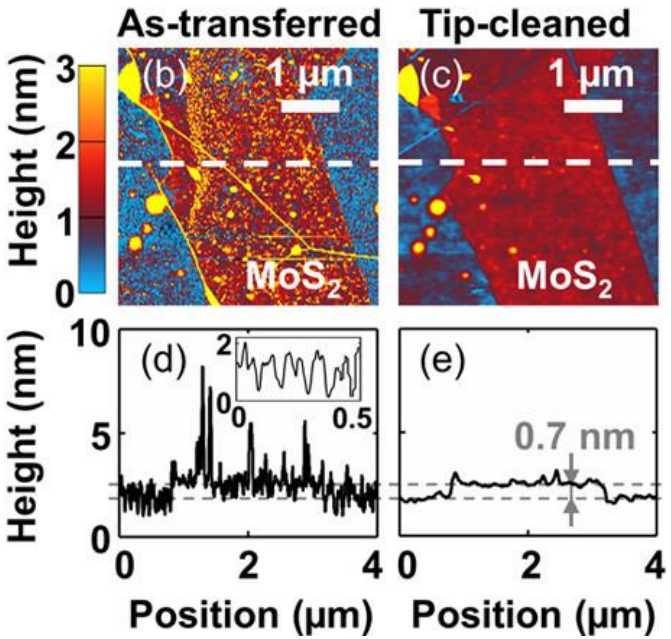
2. AFM bubble and contamnimant cleaning





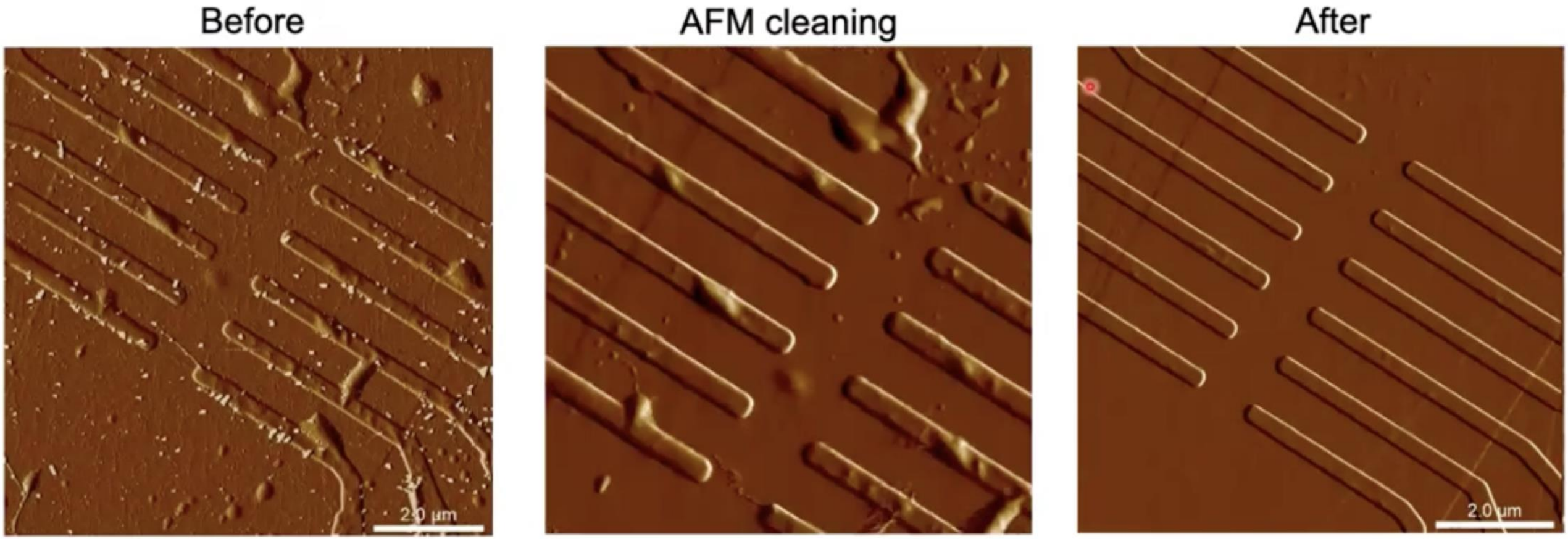
Nano sqeeze tricks





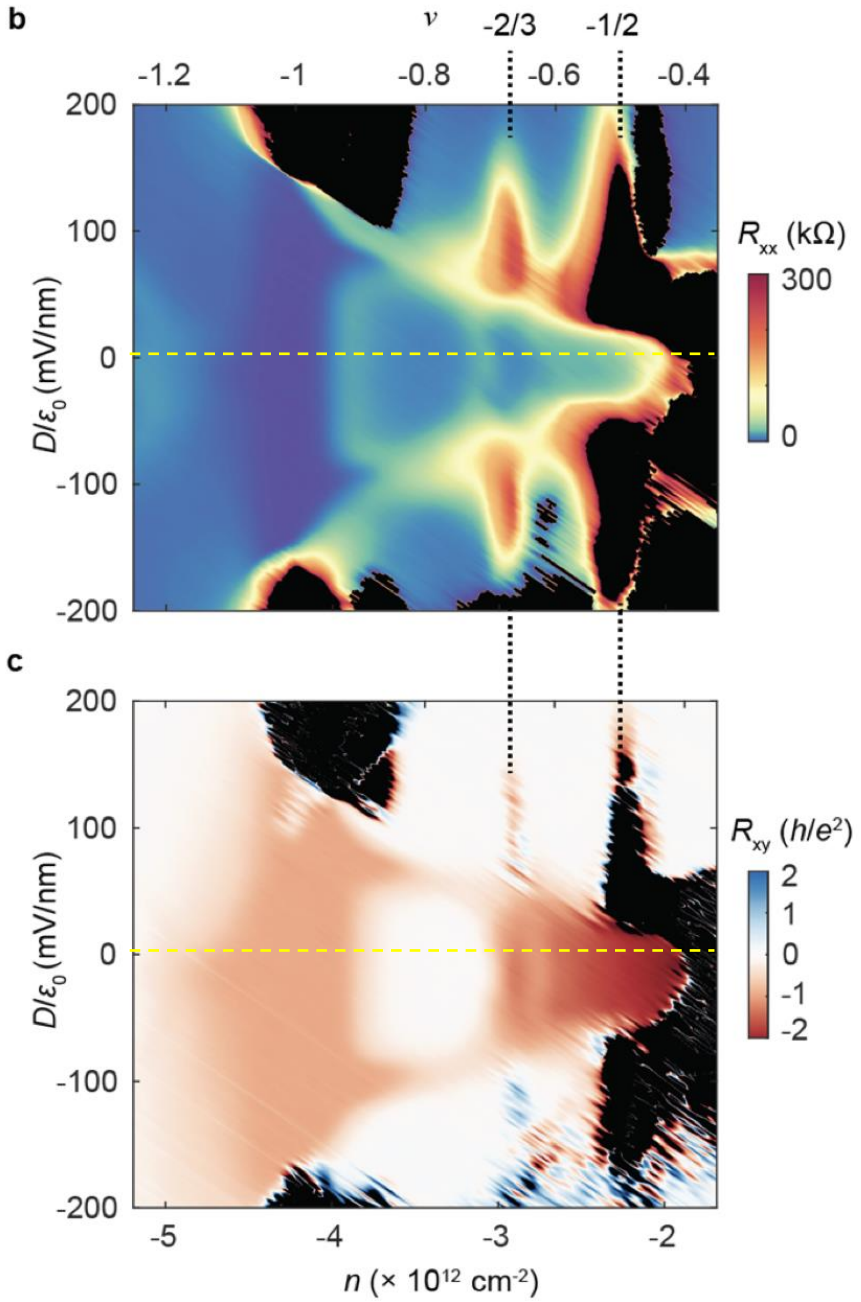
2019ACS Applied Mat. & Interf., Berend T. Jonker et al.
2021 ACS Omega, William P. King et al.

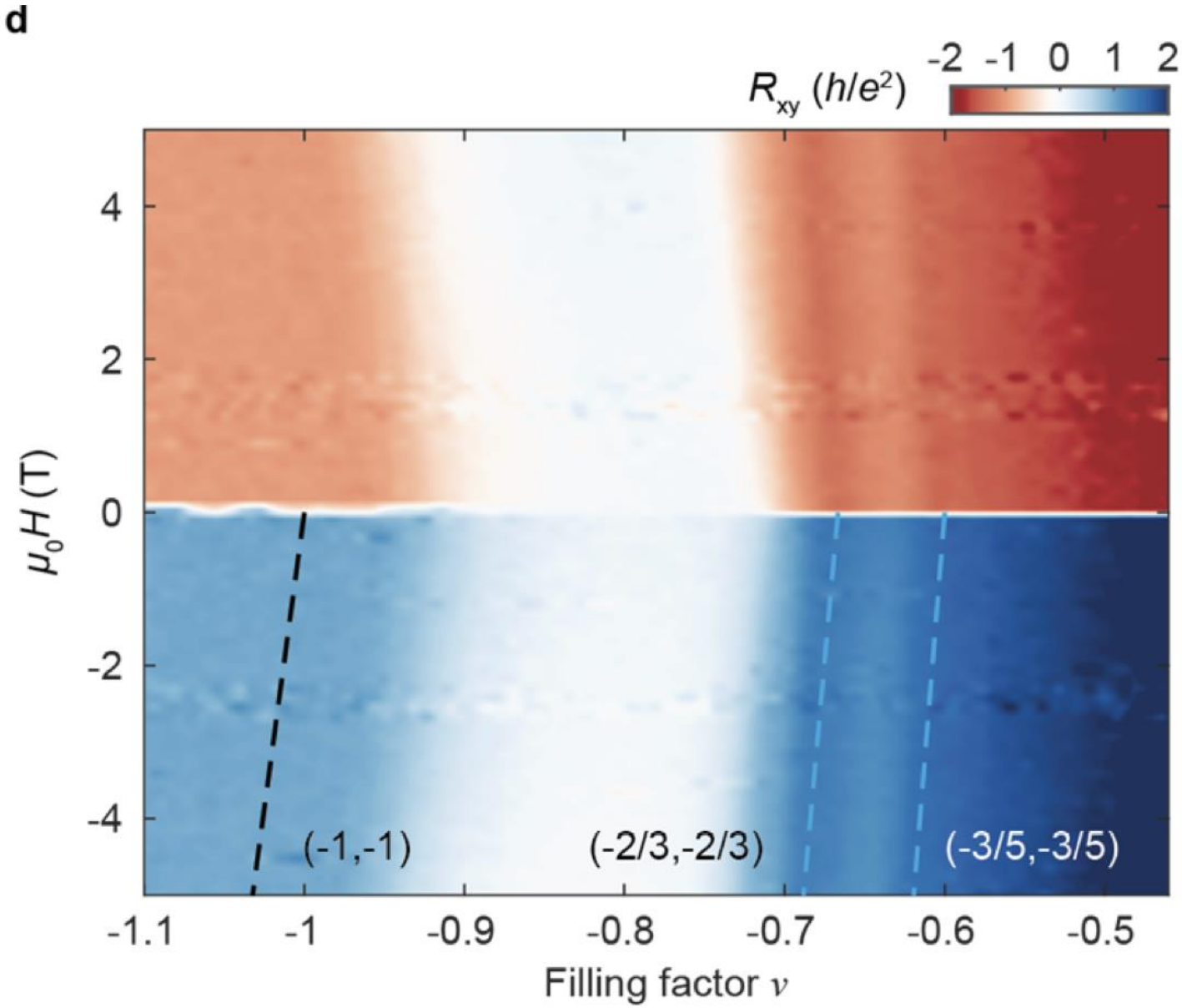
Transport device to confirm FQAH in $tMoTe_2$



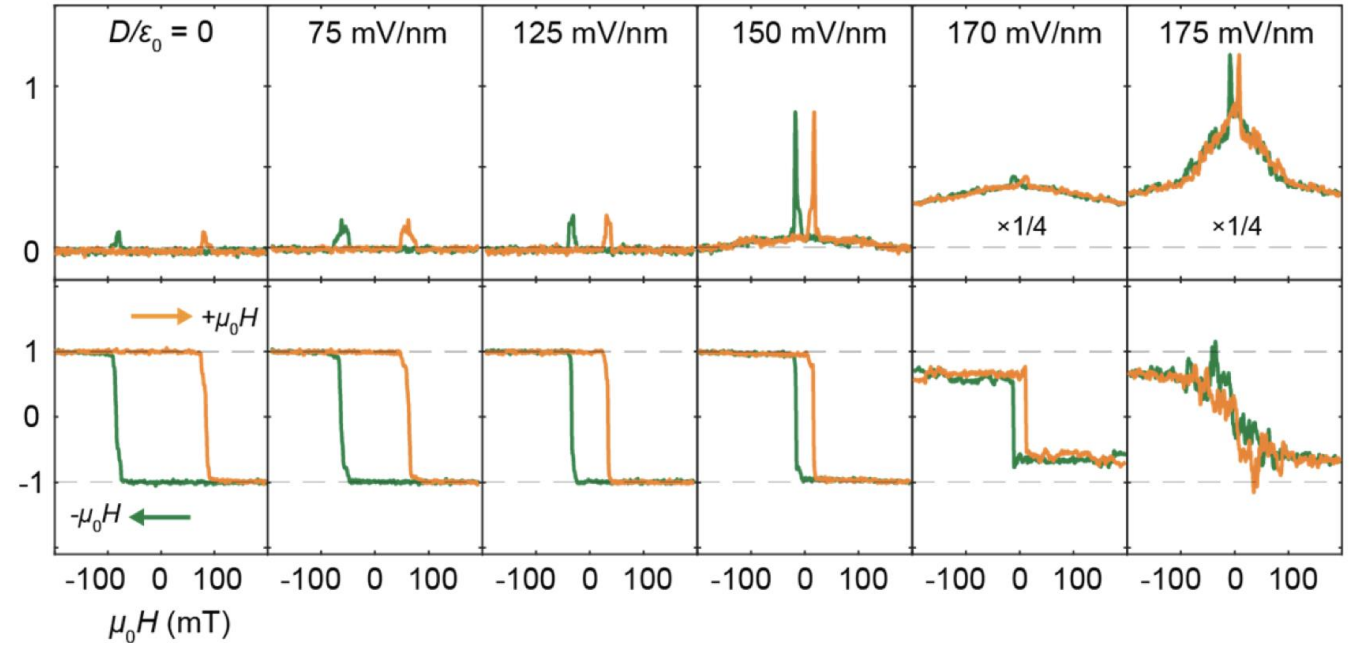
2019ACS Applied Mat. & Interf., Berend T. Jonker et al.
2021 ACS Omega, William P. King et al.

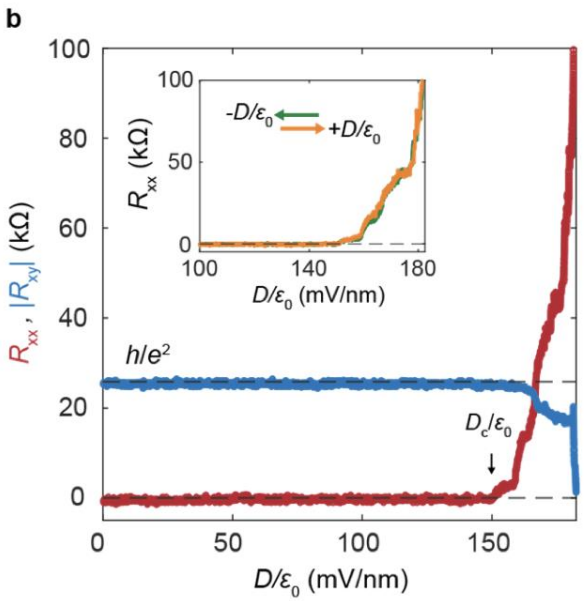
Transport device to confirm FQAH in $tMoTe_2$

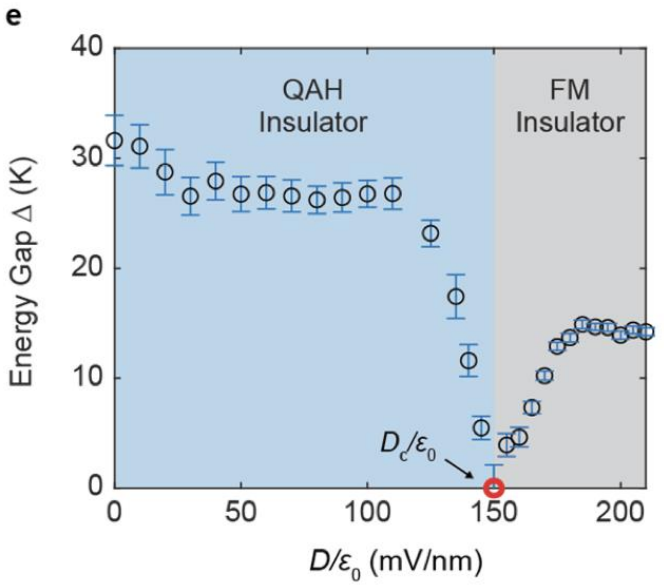




Transport of C=-1 IQAH state in $tMoTe_2$

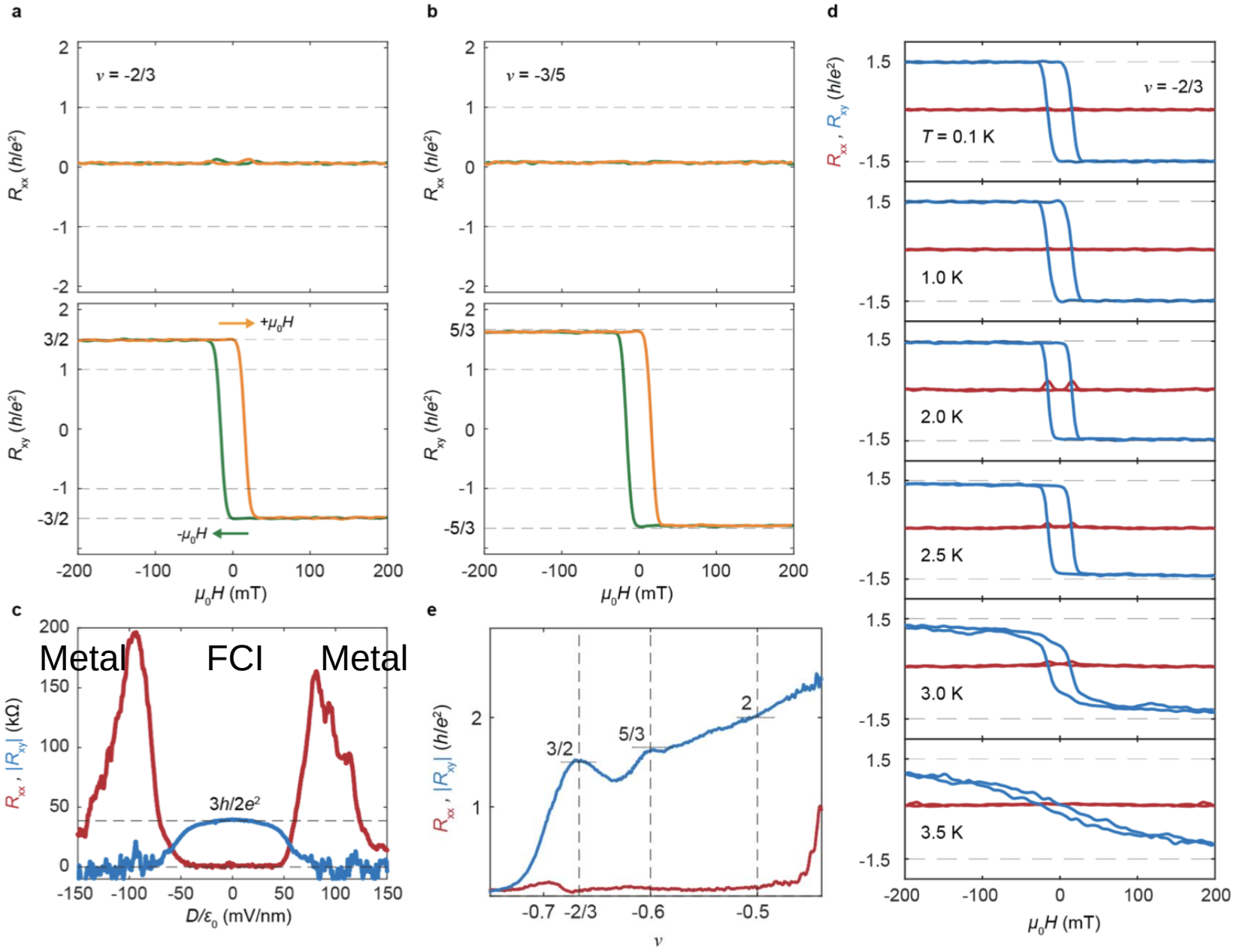






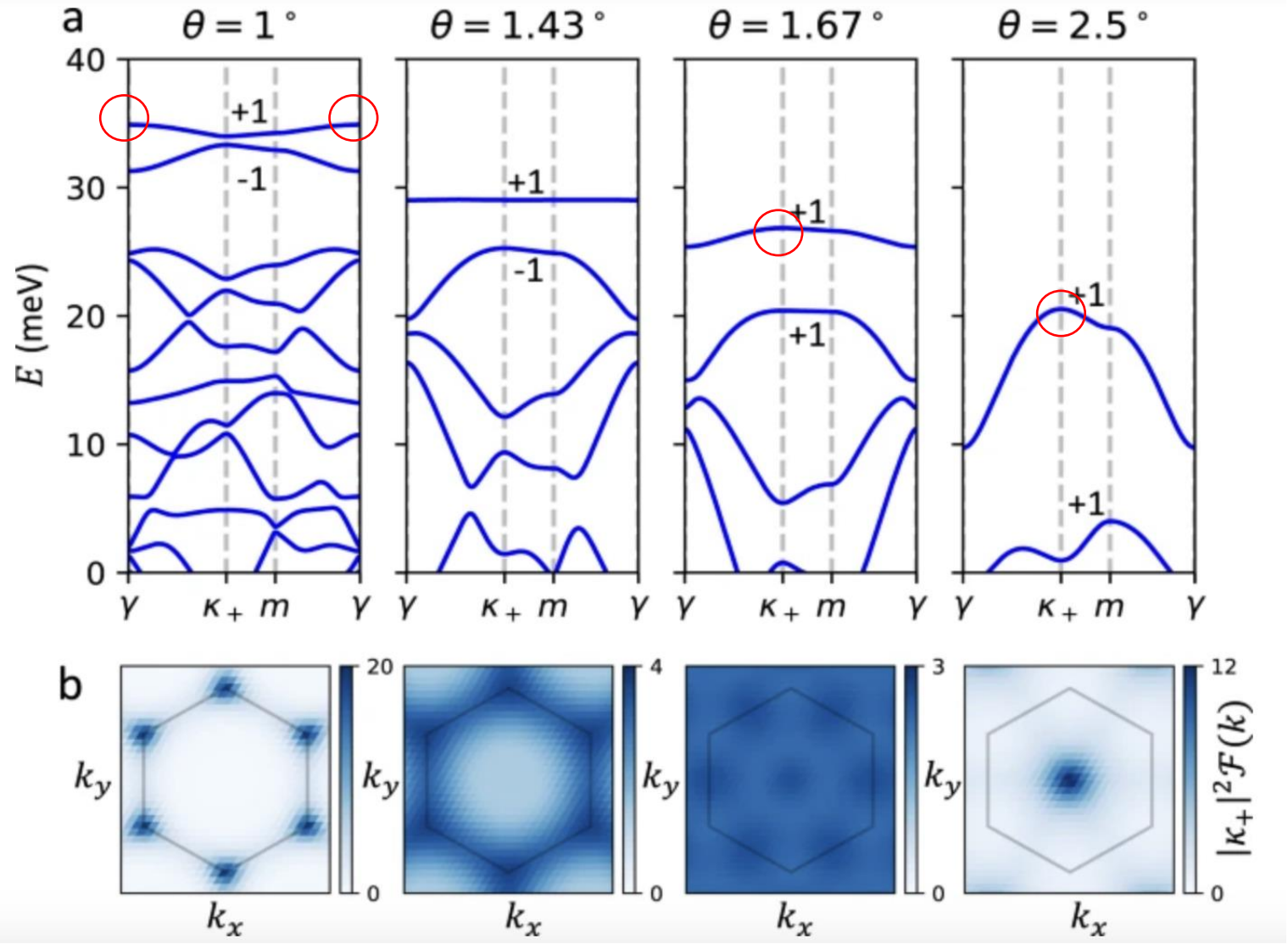
Electrical tunability
QAHI--FMI transition
Curie Temperature: 14K
Charge Gap: 28K

Transport of C=-2/3 FQAH state in $tMoTe_2$

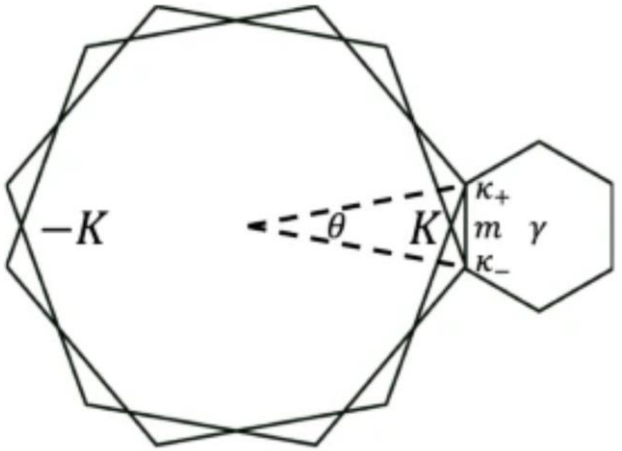


Electrical tunability
Curie Temperature: 4K
Charge Gap: 21K

$tWSe_2$ Valence band structure of mBZ under different twist -- magic angle~1.43°



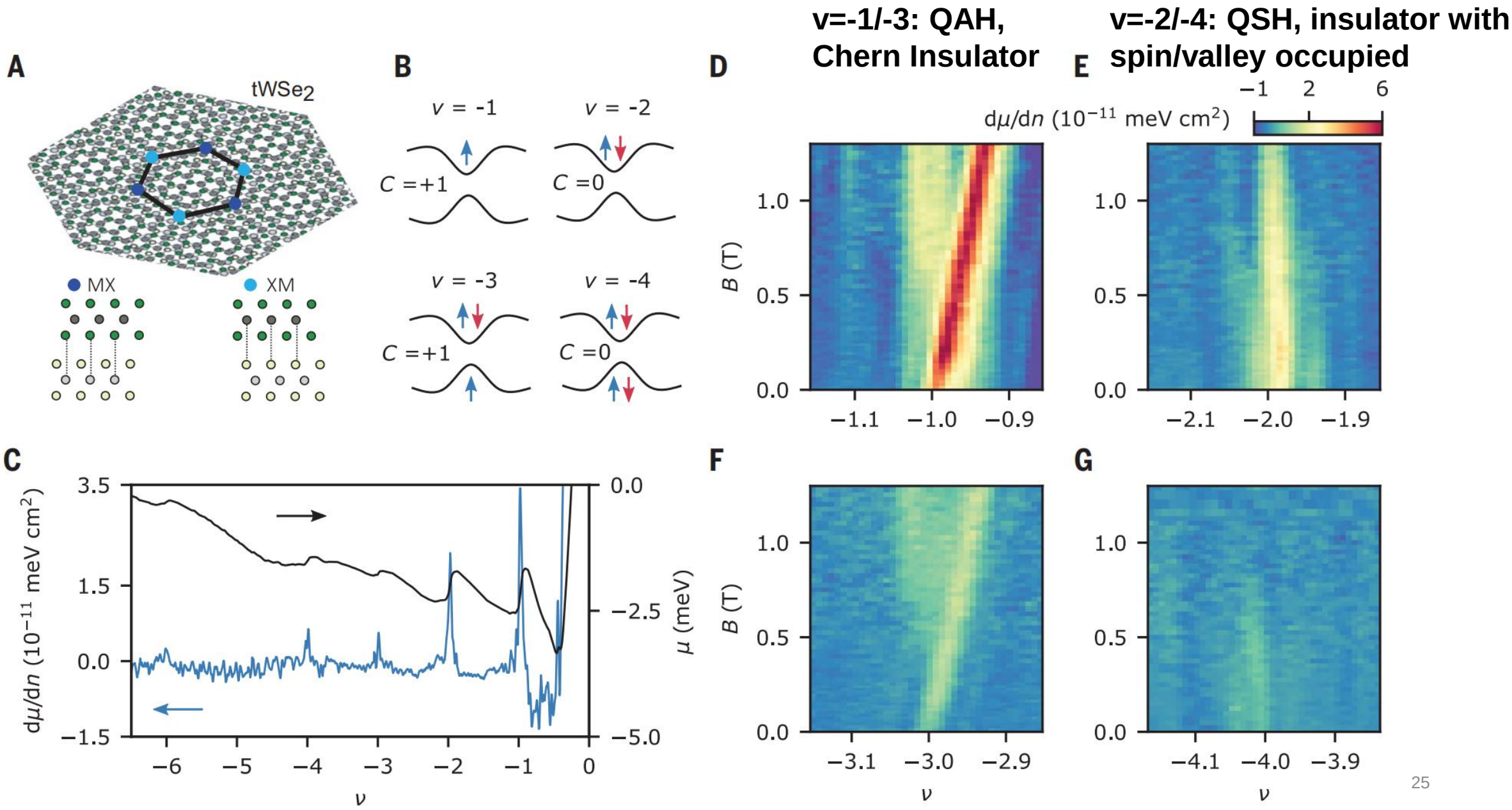
weight of top flat band wavefunction



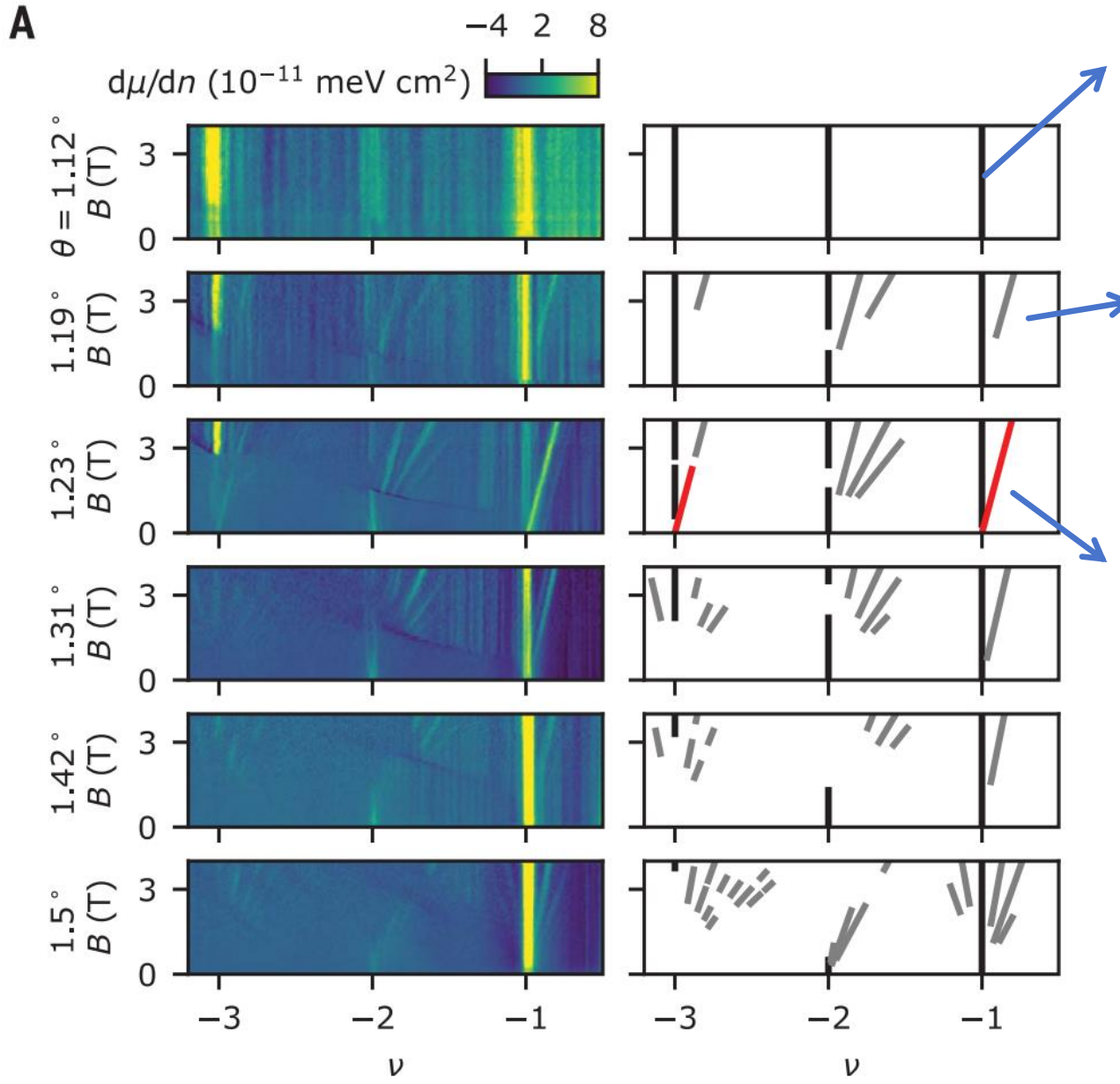
1. *γ v.s.κ±*
2. *the C=1 band is well seperated*
3. Homogenious flat Berry Curvature around 1.5°

Nat. Comm., 2021, Liang Fu

Multiple topological bands in $tWSe_2$ by scanning SET



Twist-angle dependence of correlated states and Chern insulators



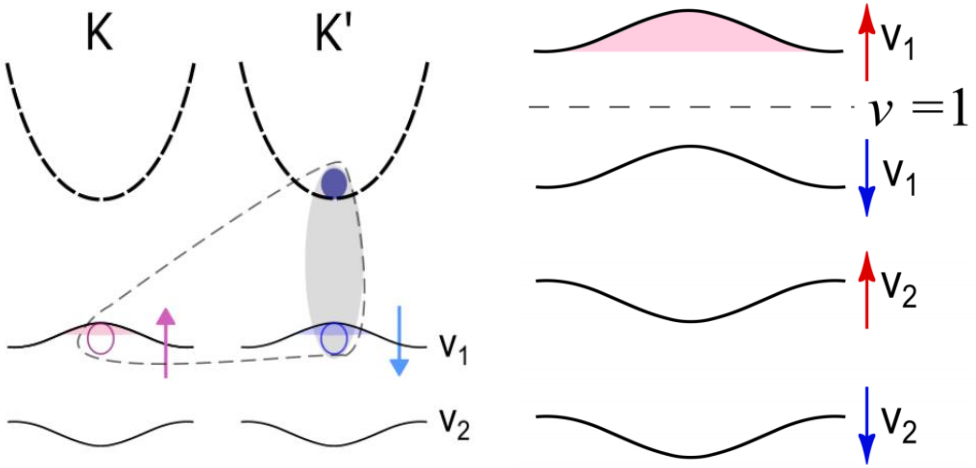
C=0

C≠0, but not persist to B=0
(Hofstadter states)

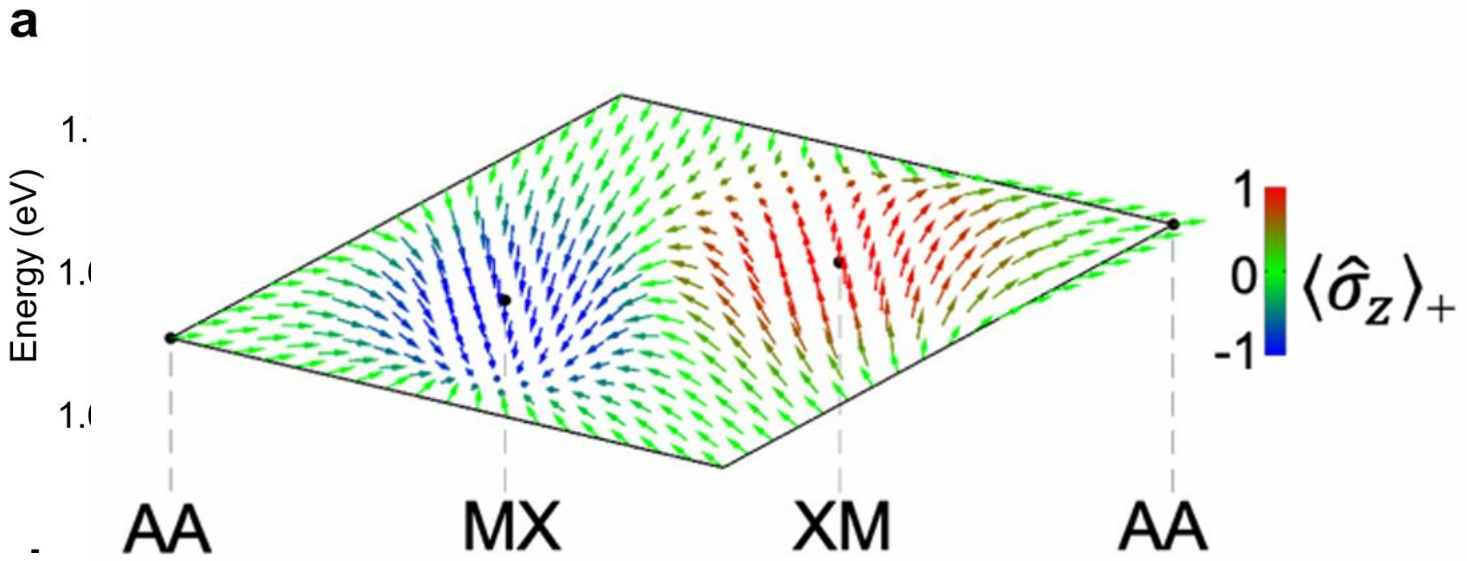
C≠0, also persist to B=0
Chern Insulators (IQAH)

QAH via attractive polaron spectroscopy with twisted angle near 2°

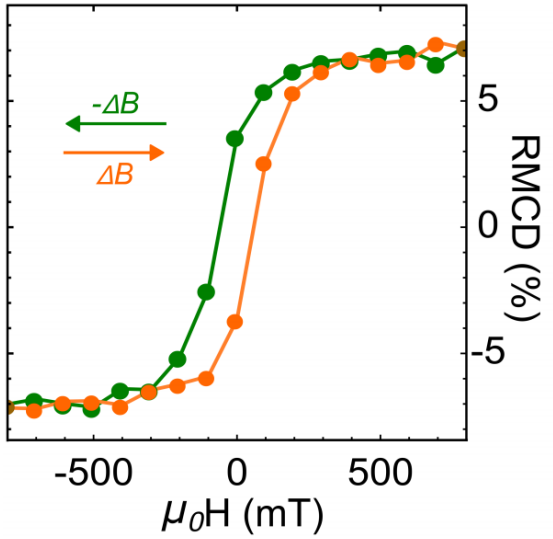
Attractive polaron



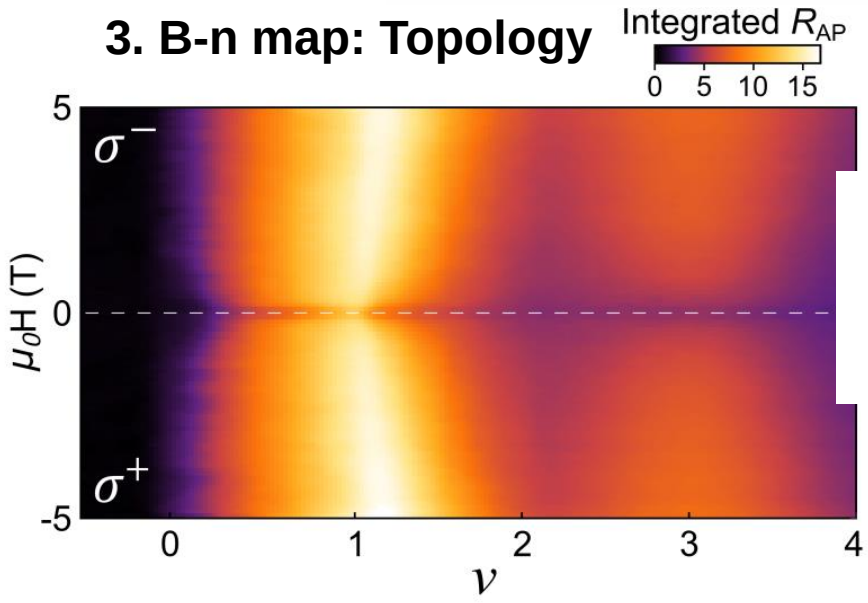
1. RC: Integer correlated states

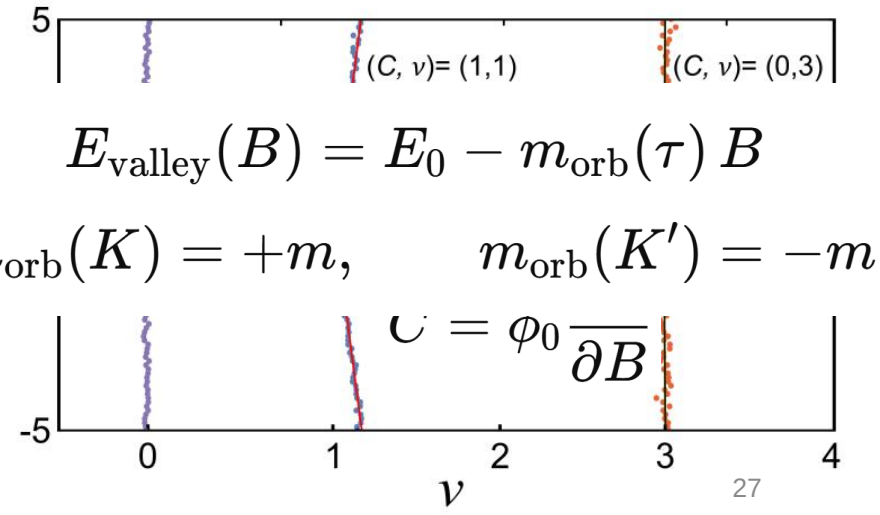


2. RMCD: Magnetism(66mK)



3. B-n map: Topology





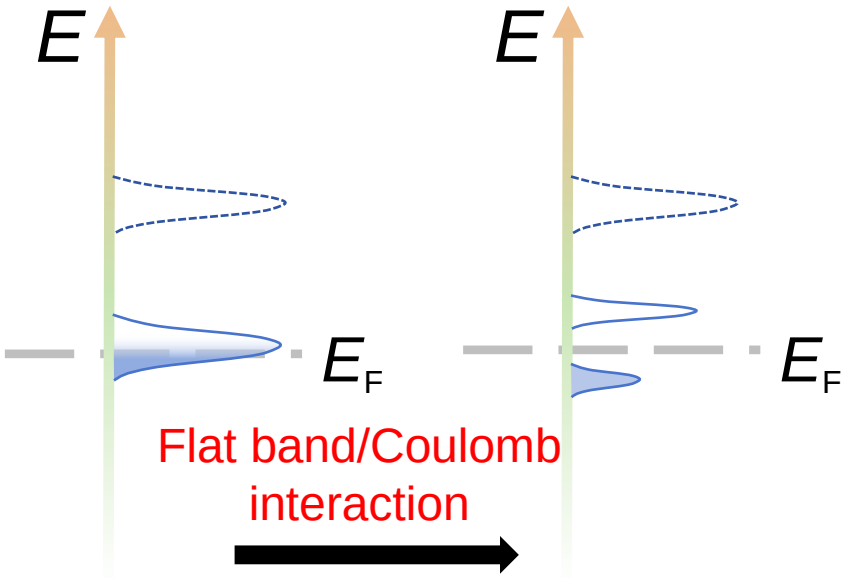
$$E_{\text{valley}}(B) = E_0 - m_{\text{orb}}(\tau)\, B$$

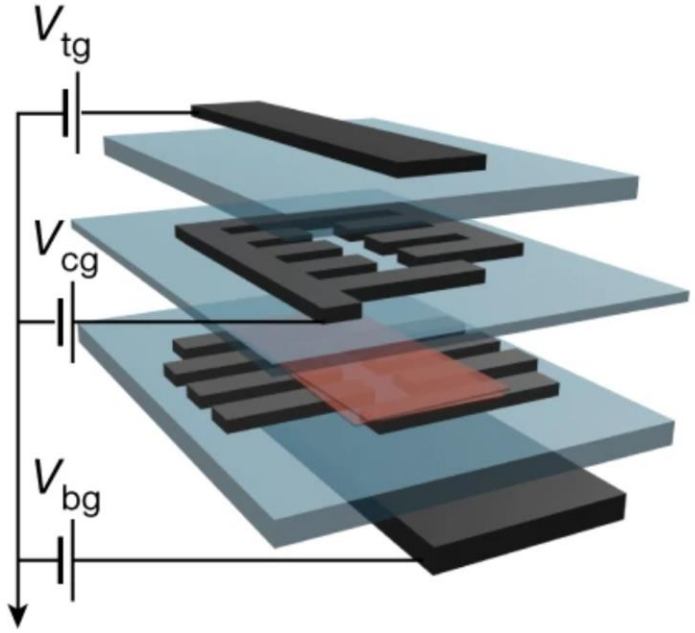
$$m_{\text{orb}}(K) = +m, \qquad m_{\text{orb}}(K') = -m$$

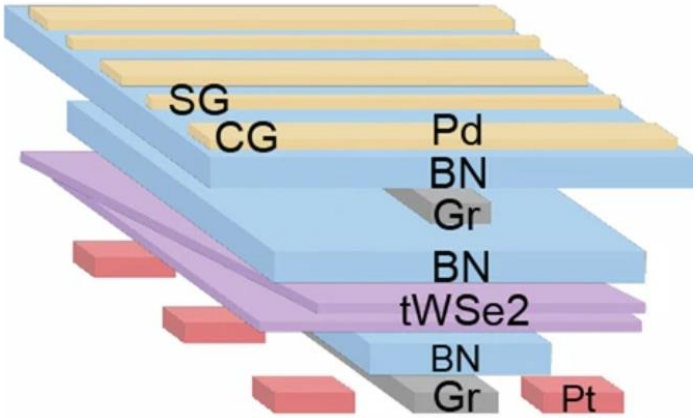
$$C = \varphi_0 \frac{}{\partial B}$$

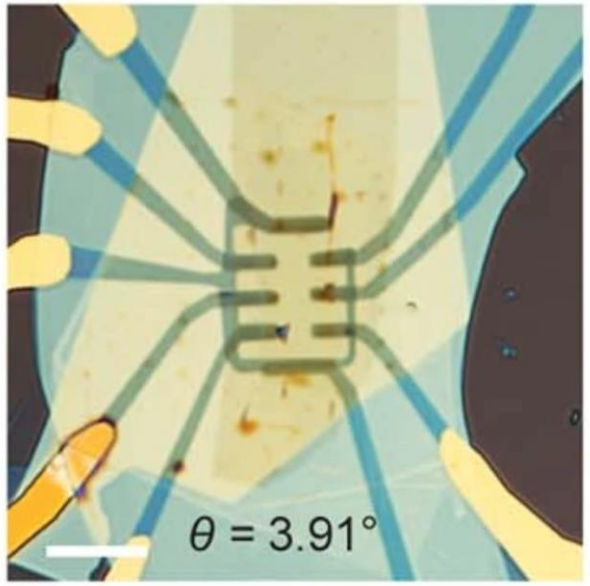
How to generate Fractional States in $tWSe_2$ based on topology?

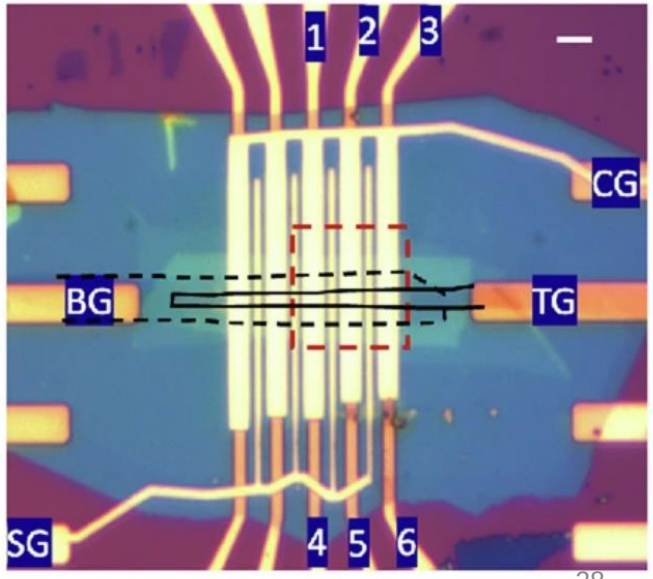
Physics: what is still not discovered in tWSe2? ***FQH/FQAH states***











S.I.

Background: Fractional Chern Insulators (FCIs)

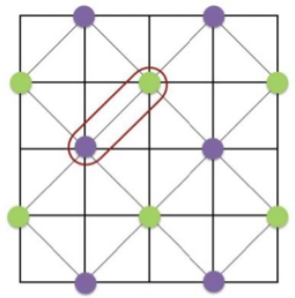
- 2-D electronic systems which realize FQH states onto a lattice without external magnetic field.

- Flat band dispersion simulate Landau levels: high flatness ratio

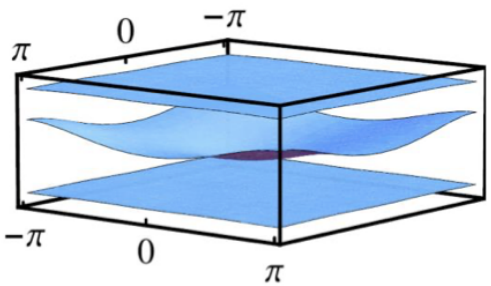
$$r = \frac{bandgap}{bandwidth}$$

- Uniform (flat) Berry curvature F_{xy} simulate uniform magnetic field

- Interaction term with desired locality in FQH basis



CB Model



One of the first FCI models devised: The Checkerboard (CB) model with flatness ratio ~ 30. (Kai Sun et. al. 2011)

→ Ideal Situation

AB and AC effect

VOLUME 53, NUMBER 4 PHYSICAL REVIEW LETTERS 23 JULY 1984

Topological Quantum Effects for Neutral Particles

Y. Aharonov[(a)] and A. Casher
Department of Physics and Astronomy, Tel Aviv University, Tel Aviv 69978, Israel
(Received 21 May 1984)

We derive the effective Lagrangian which describes the interaction between a charged particle and a magnetic moment in the nonrelativistic limit. It is shown that neutral particles with a magnetic moment will exhibit the Aharonov-Bohm effect in certain circumstances. We suggest several types of experiments.

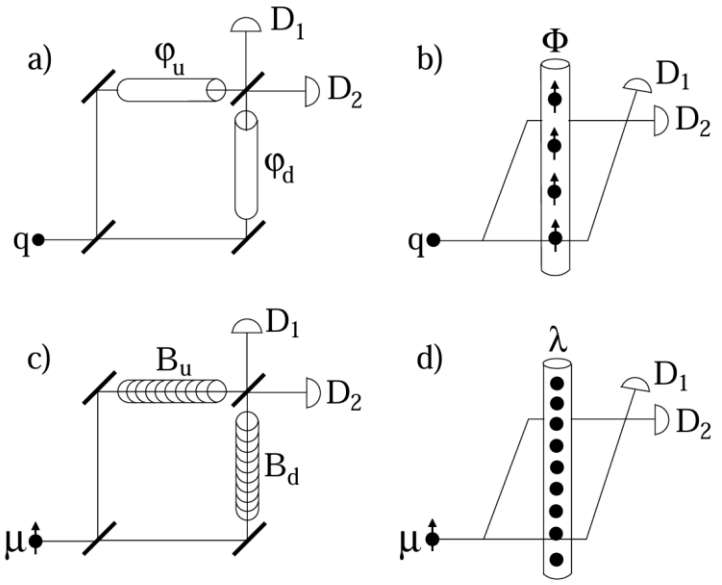
PACS numbers: 03.65.Bz

For charged particles (Fermions):

$$\Delta\varphi \sim \frac{q}{\hbar c}\oint \mathbf{A}\cdot d\mathbf{r} \qquad \Delta\phi_{\mathrm{AB}} = \frac{q\Phi}{\hbar c}$$

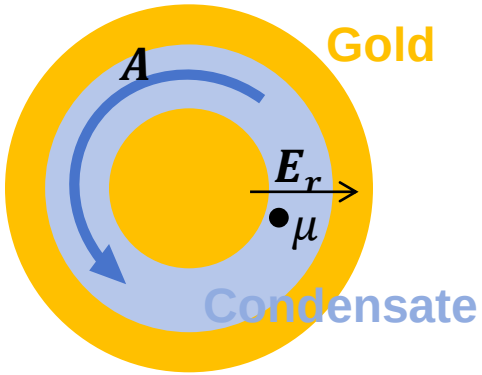
For neutral particles (Excitons):

$$\Delta\varphi \sim \frac{1}{\hbar c^2}\oint(\boldsymbol{\mu}\times\mathbf{E})\cdot d\mathbf{r} \qquad \Delta\phi_{\mathrm{AC}} = \frac{4\pi\lambda\mu_{\parallel}}{\hbar c^2}$$



$$I(\Phi) \propto 1 + \cos\left(\frac{q\Phi}{\hbar c}\right)$$

AC effect in Corbino structure



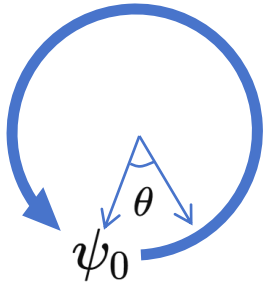
Effective vector field:

$$\oint \mathbf{A} \cdot d\mathbf{r}$$

For neutral particles (Excitons):

$$\Delta\phi_{\mathrm{AC}} = \frac{1}{\hbar c^2} \oint (\boldsymbol{\mu} \times \mathbf{E}) \cdot d\mathbf{r}$$

For a neutral ring condensate with global coherence (phase stiffness/U(1)):



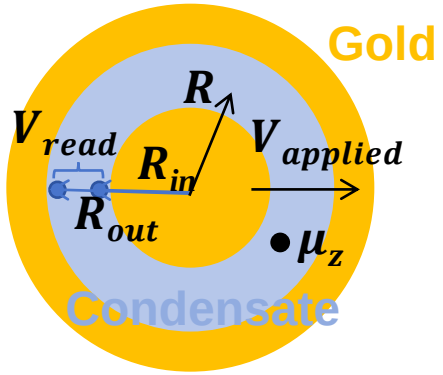
$$\psi(\theta) = \psi_0\, e^{i\phi(\theta)}$$

$$\psi_0\, e^{i\phi(\theta+2\pi)} = \psi_0\, e^{i\phi(\theta)}$$

$$e^{i[\phi(\theta+2\pi)-\phi(\theta)]} = 1 \quad \phi(\theta+2\pi) - \phi(\theta) = 2\pi n$$

$$\Delta\phi_{\mathrm{AC}} = \phi(\theta+2\pi) - \phi(\theta) = 2\pi n$$

Corbino device readout



$$V_{read} \sim n \frac{\hbar c^2 (R_{\text{outer}} - R_{\text{inner}})}{R \mu_z}$$

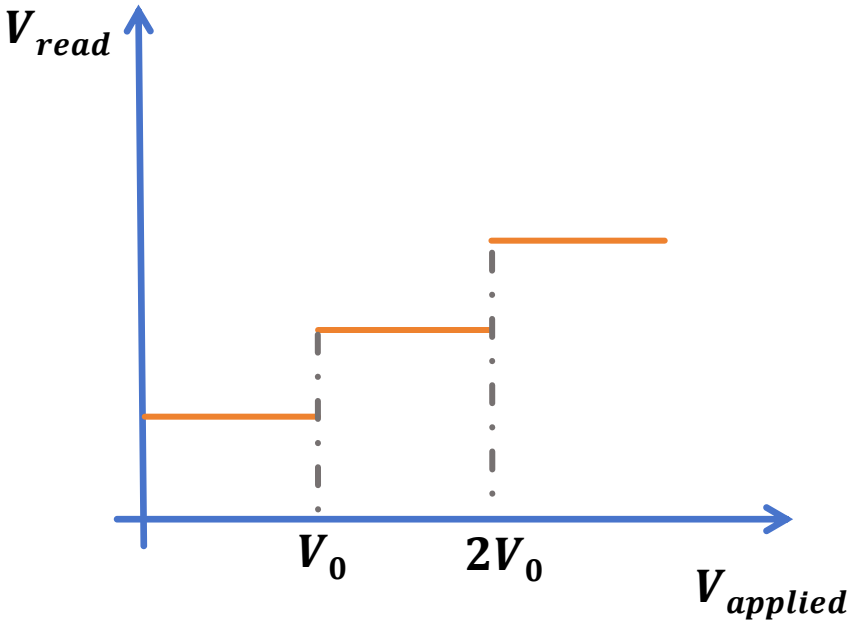
Electric Field Quantization:

$$\boldsymbol{\mu} \times \mathbf{E} = \mu_z \hat{z} \times (E_r \hat{r}) = \mu_z E_r \hat{\theta}$$

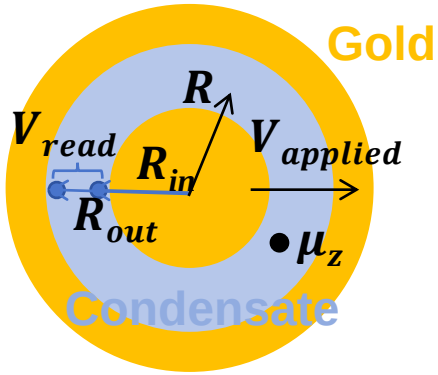
$$(\boldsymbol{\mu} \times \mathbf{E}) \cdot d\mathbf{r} = \mu_z E_r \, \hat{\theta} \cdot (R d\theta \, \hat{\theta}) = \mu_z E_r R d\theta$$

$$\Delta\phi_{\text{AC}} = \frac{1}{\hbar c^2} \int_0^{2\pi} \mu_z E_r R \, d\theta$$

$$\Delta\phi_{\text{AC}} = \frac{2\pi R \mu_z E_r}{\hbar c^2} = 2\pi n$$



What is the best recipe?



$$V_{read} \sim n \frac{\hbar c^2 (R_{outer} - R_{inner})}{R \mu_z}$$

Smaller

Larger